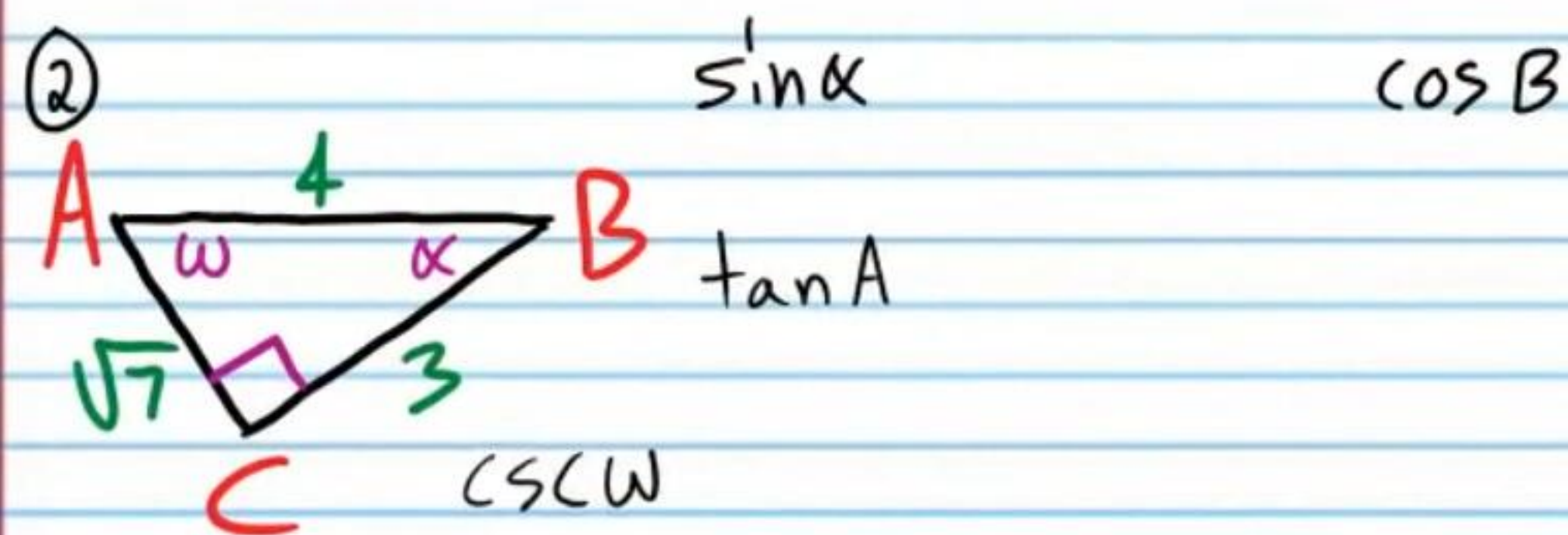
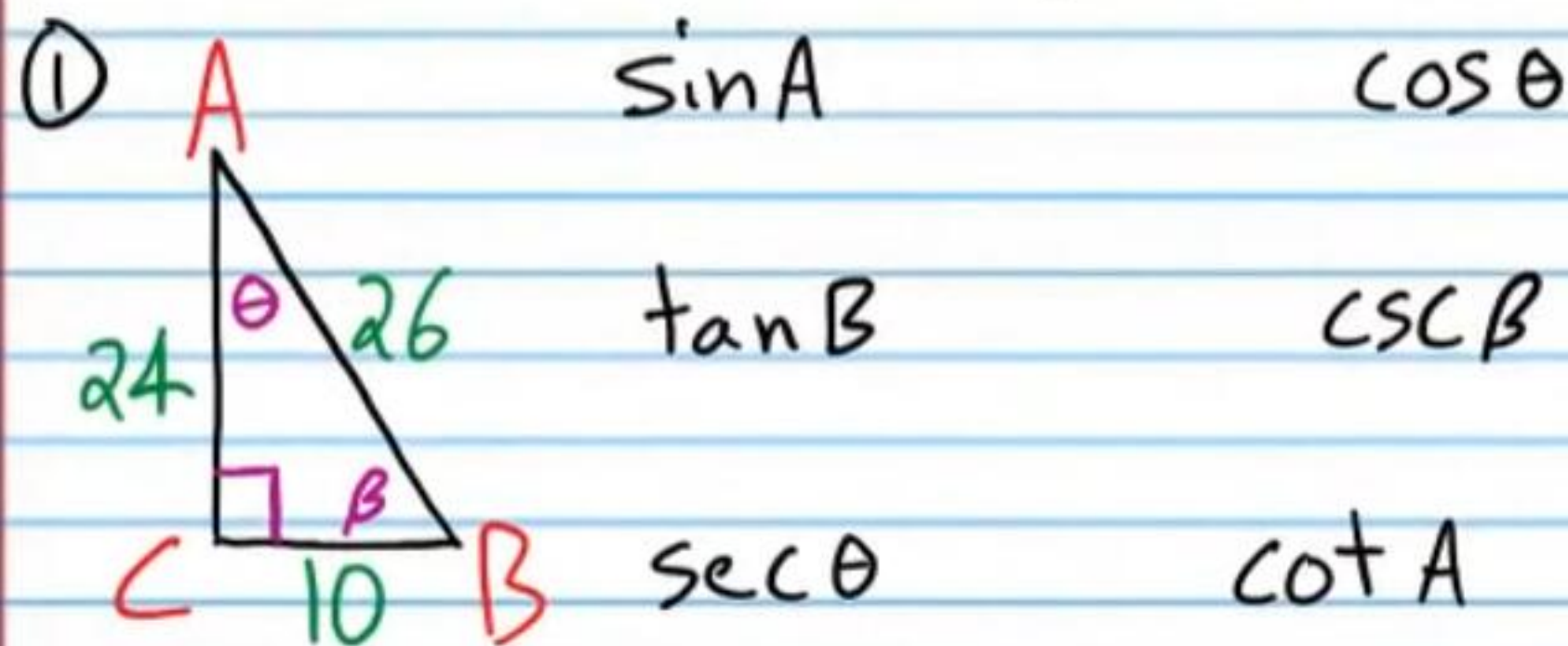


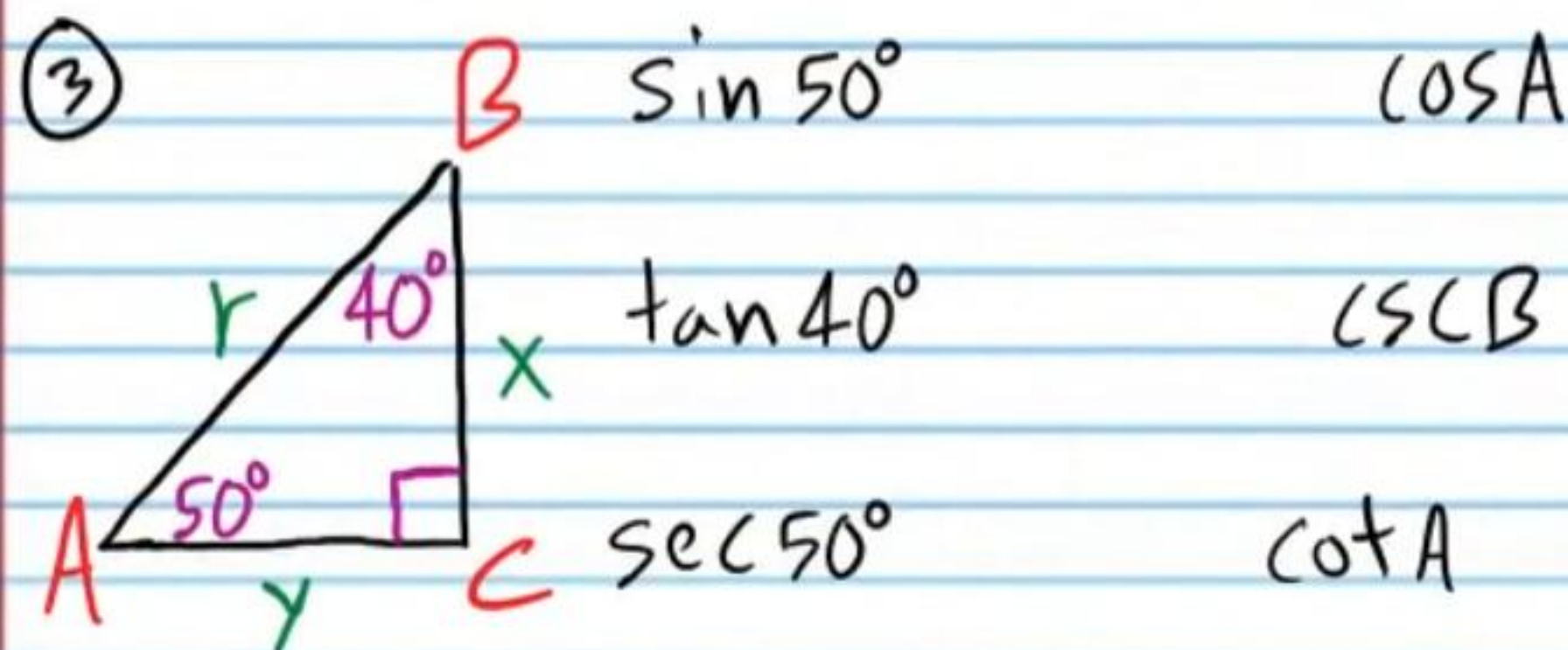
Trigonometry Review (Homework)

Date
Course

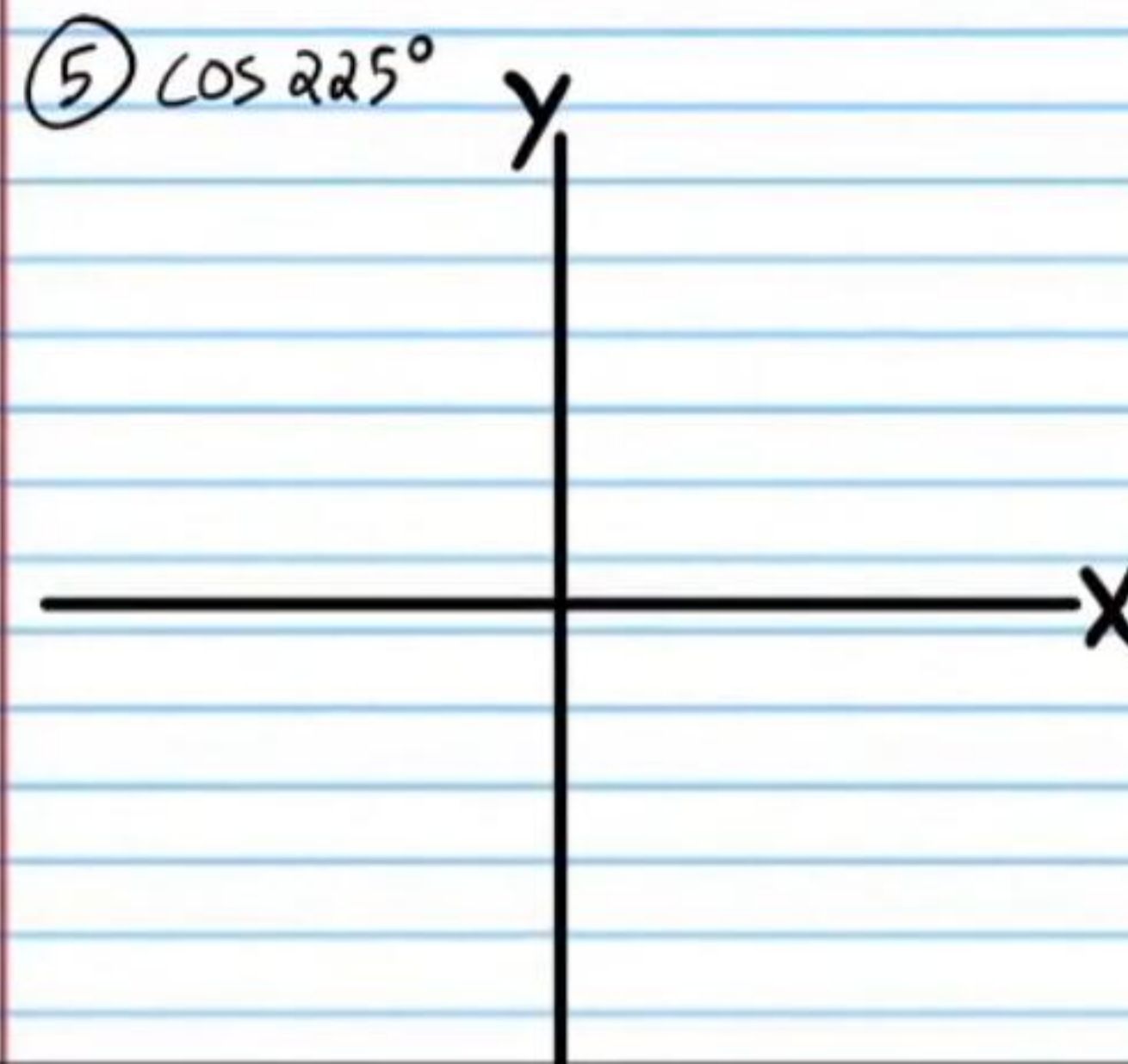
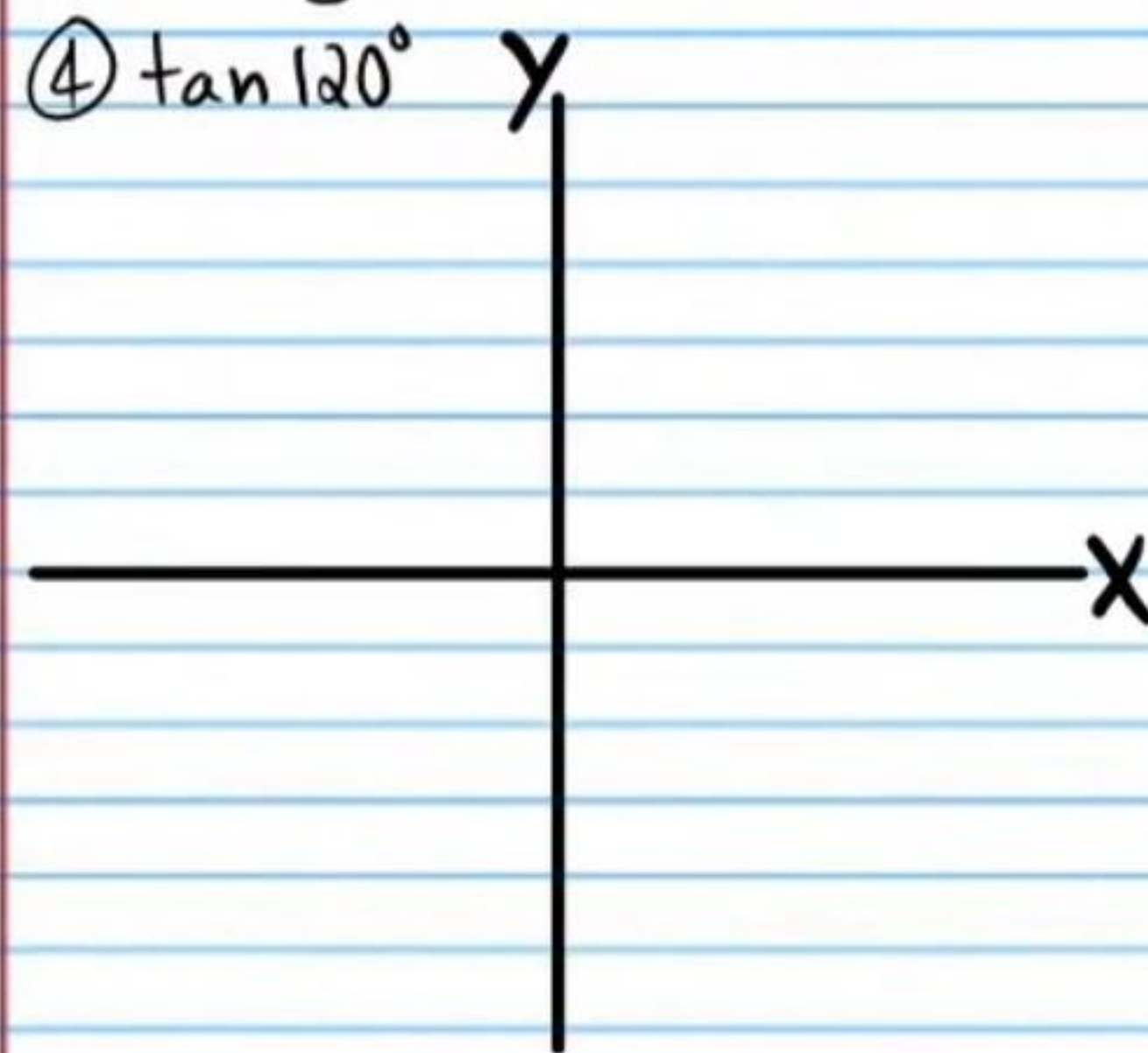
Find the values of the expressions below.



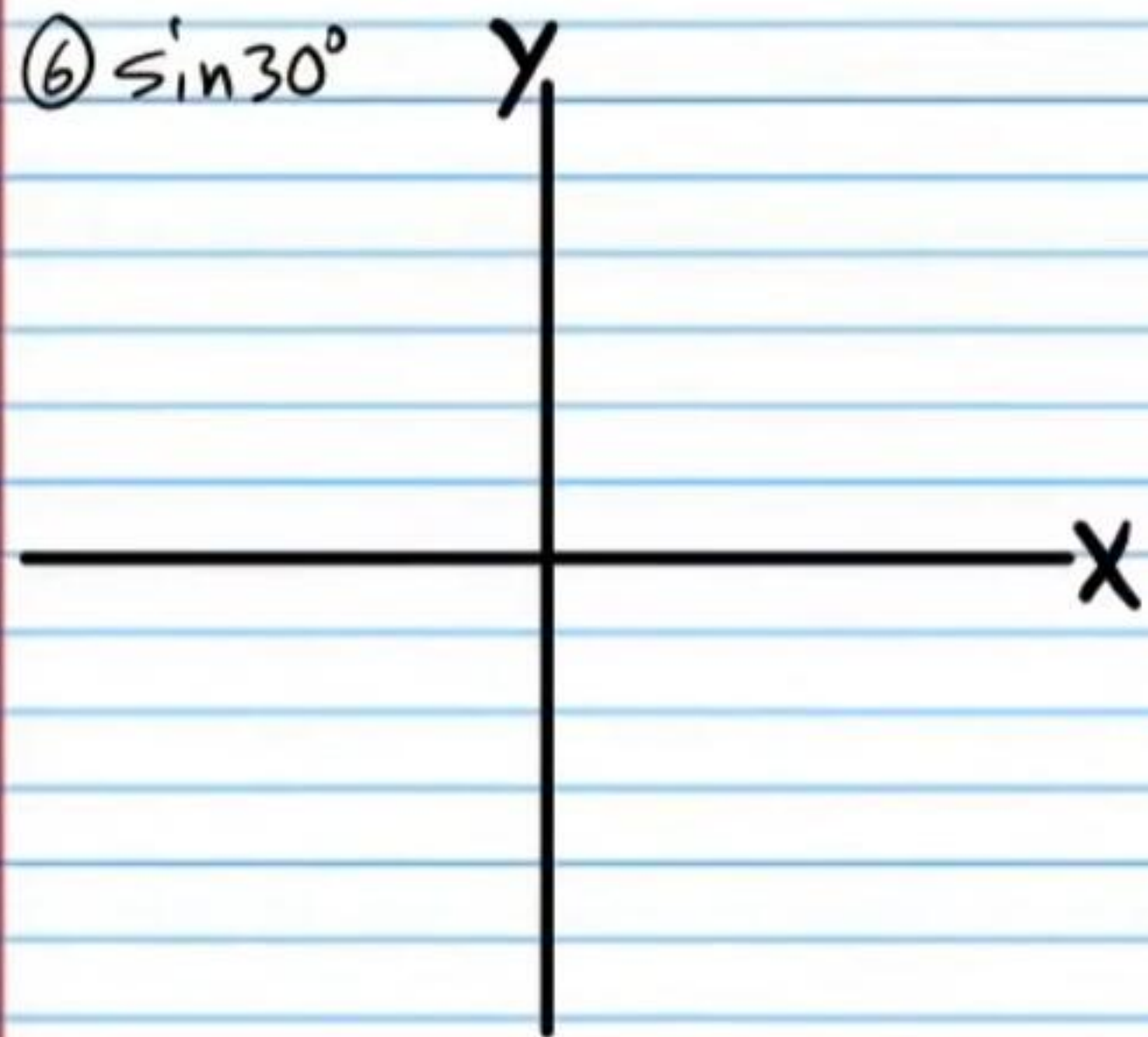
$\sec A$ $\cot B$



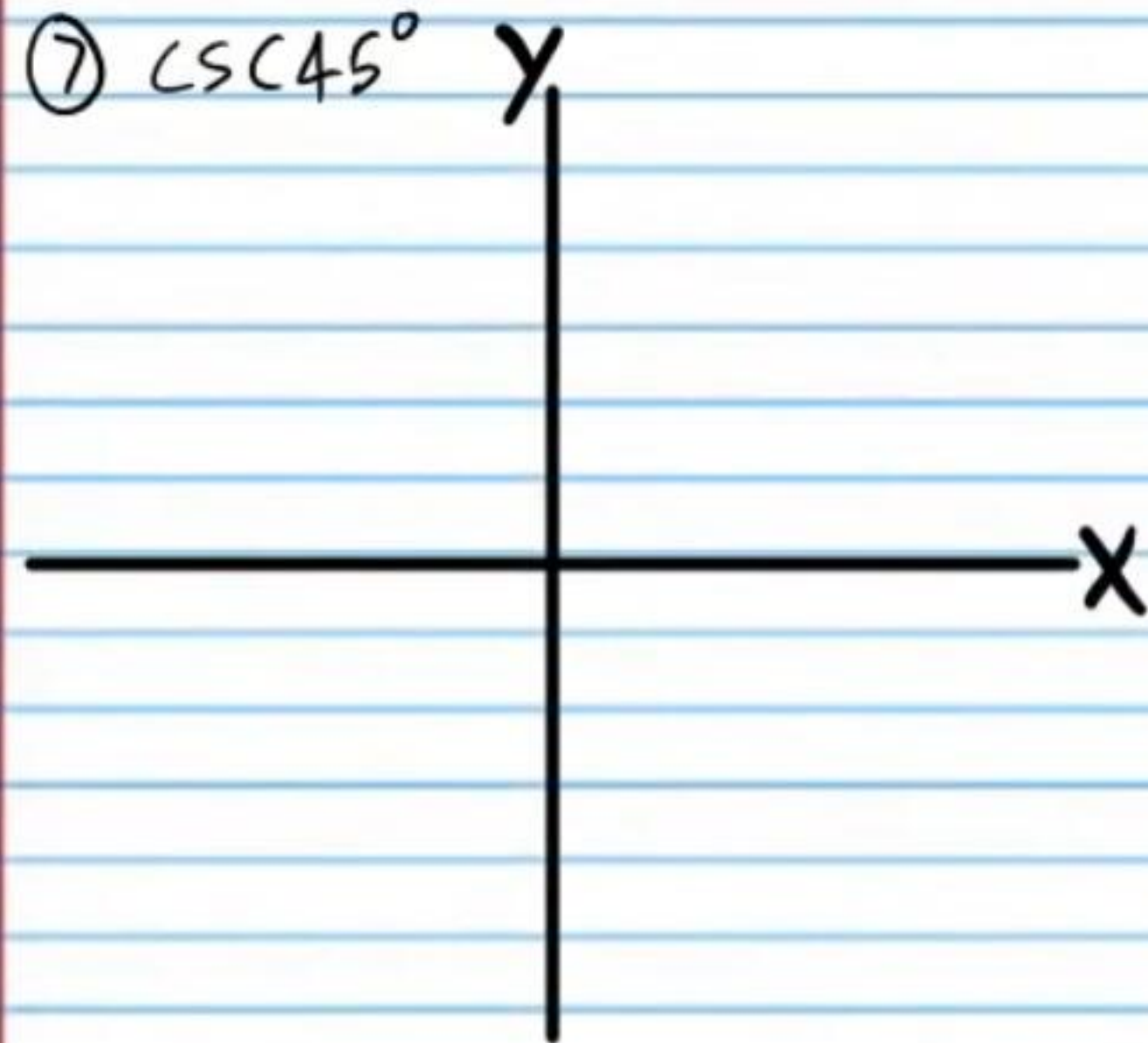
Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help find the value of the trigonometric expression. Then, find the value of the trigonometric expression *without a calculator*.



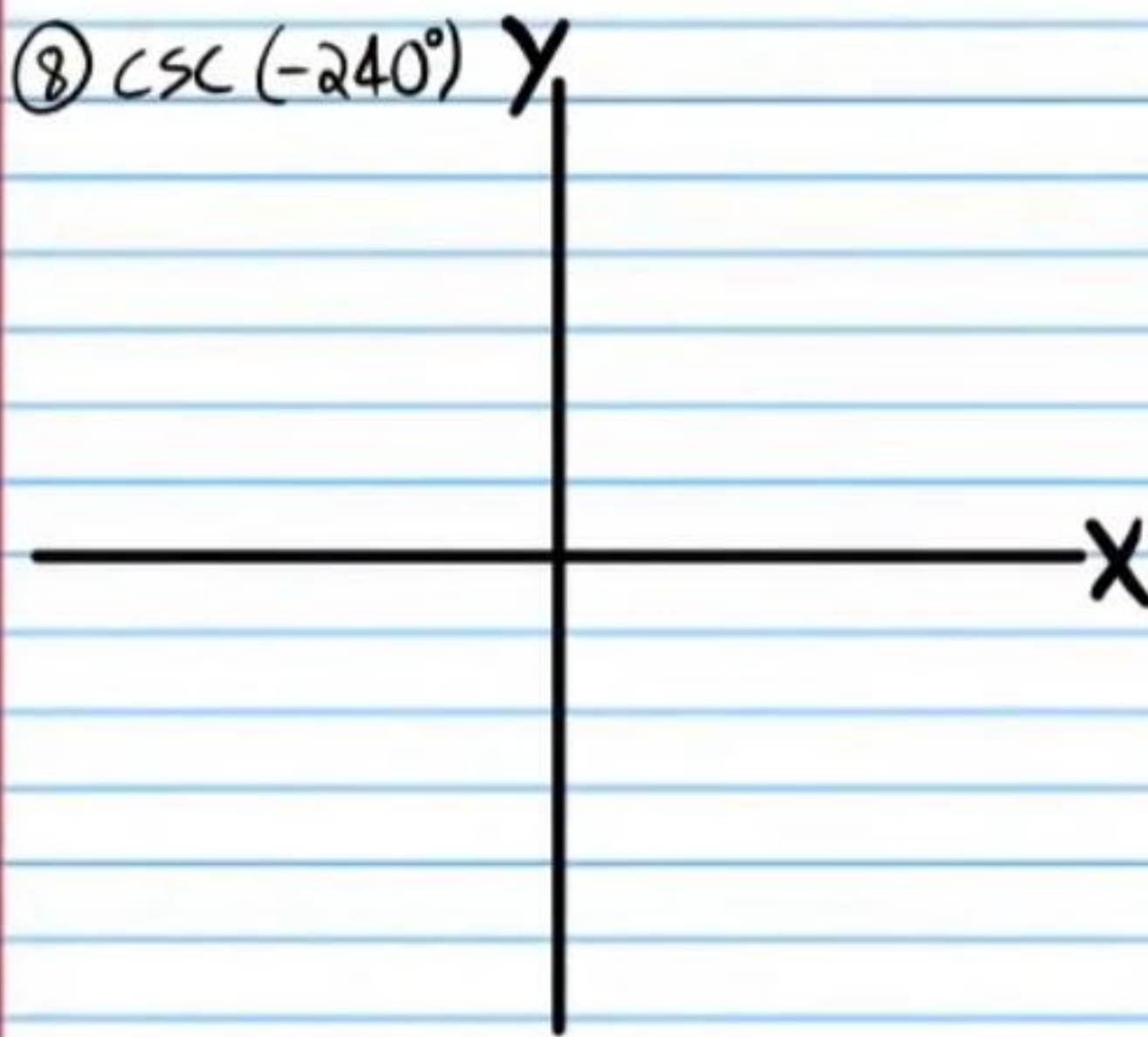
⑥ $\sin 30^\circ$



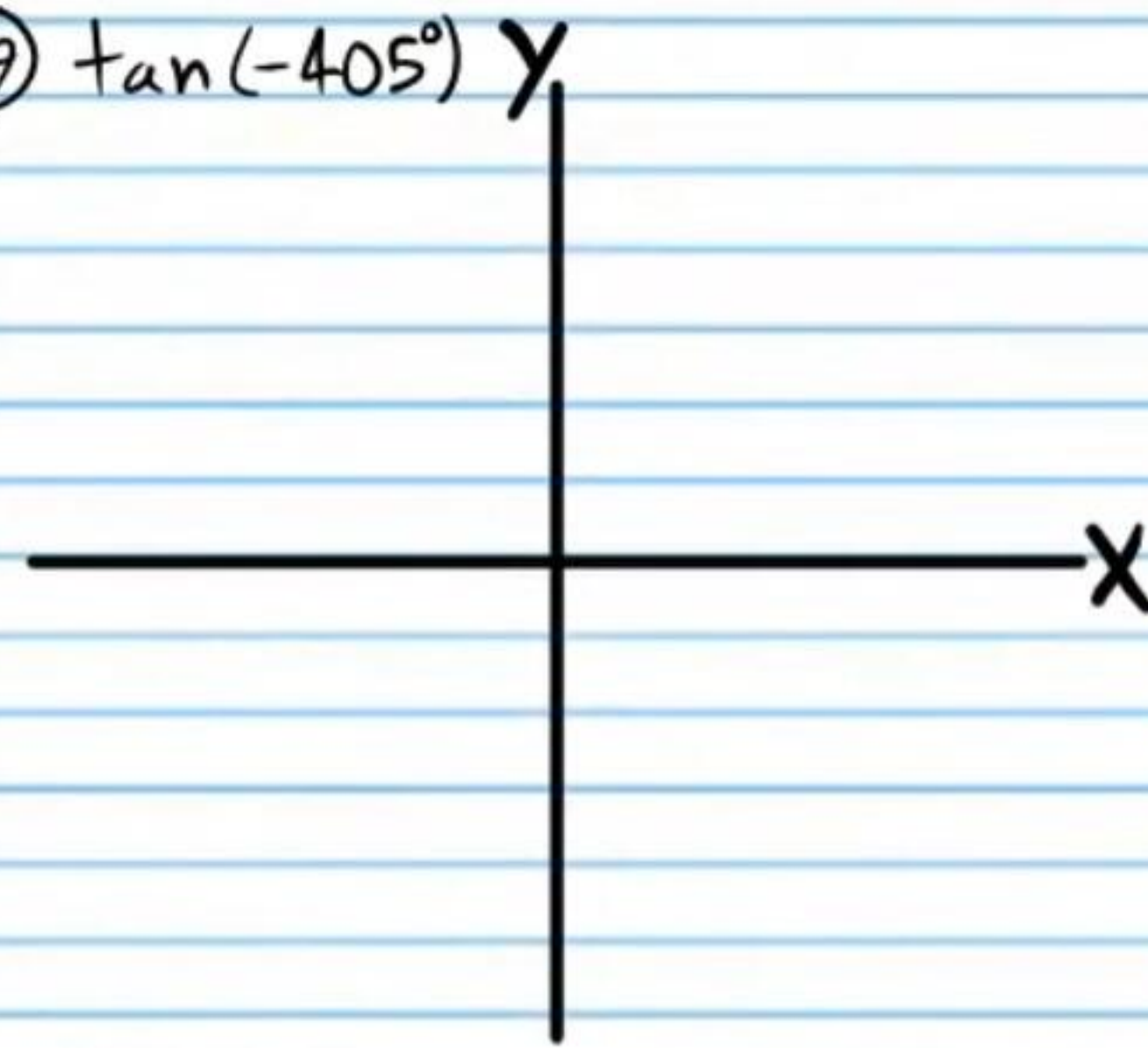
⑦ $\csc 45^\circ$



⑧ $\csc(-240^\circ)$

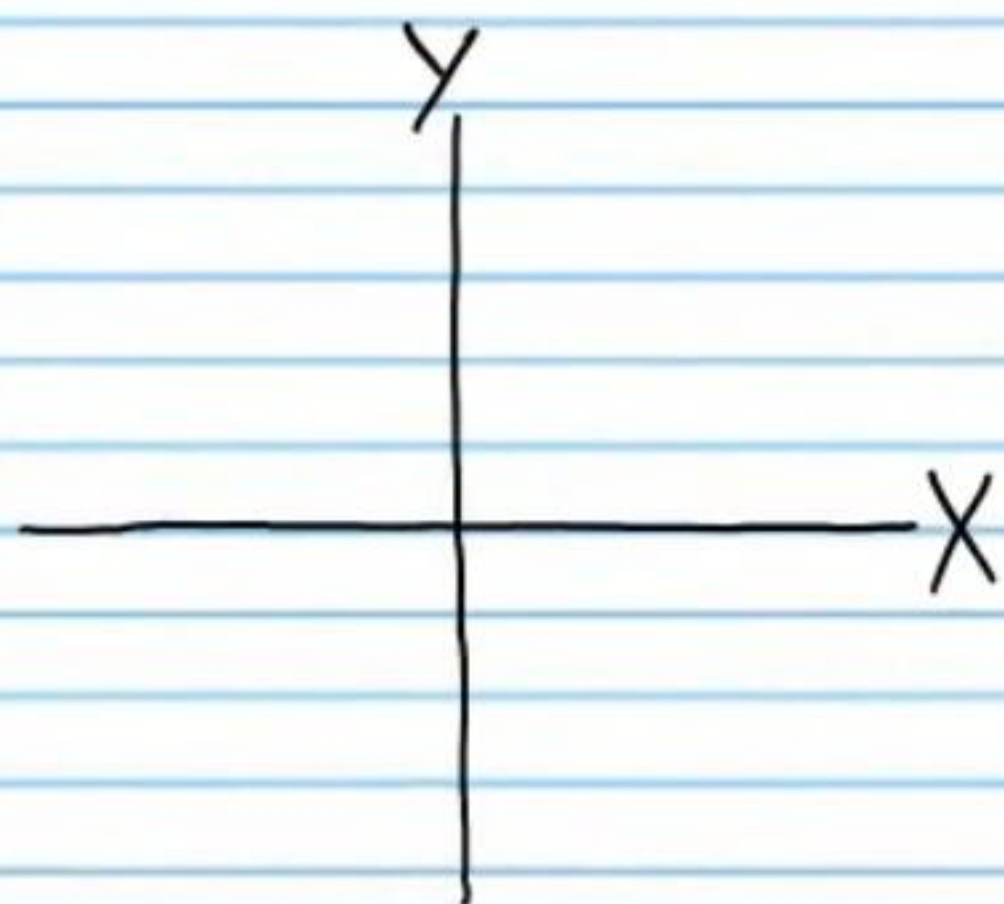


⑨ $\tan(-405^\circ)$

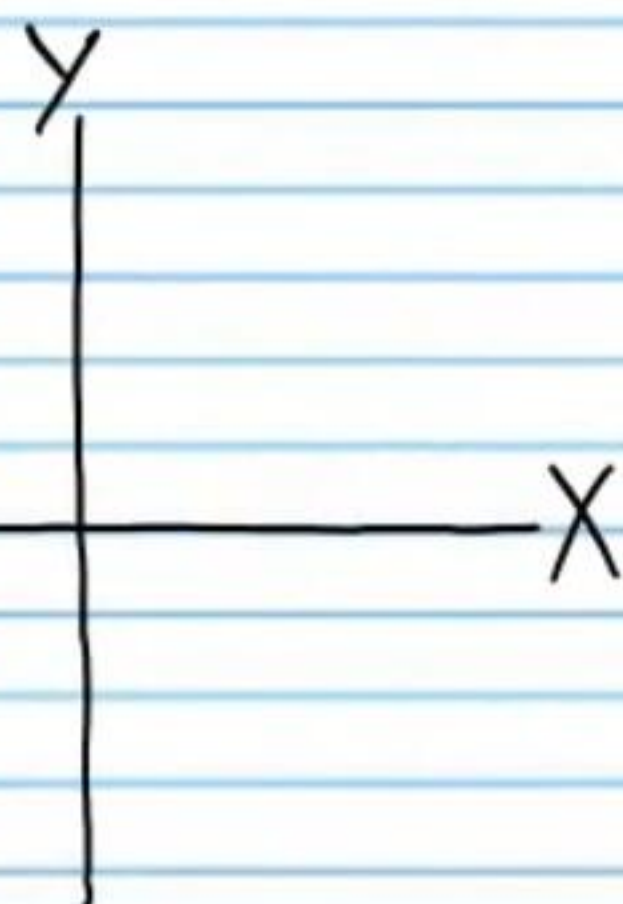


Draw each angle in standard position, write the measure of the reference angle, and draw a triangle to help understand the value of the trigonometric expression. Then, use a calculator to determine the values of each trigonometric expression. Round to four decimal places. Explain why some values are positive and some are negative.

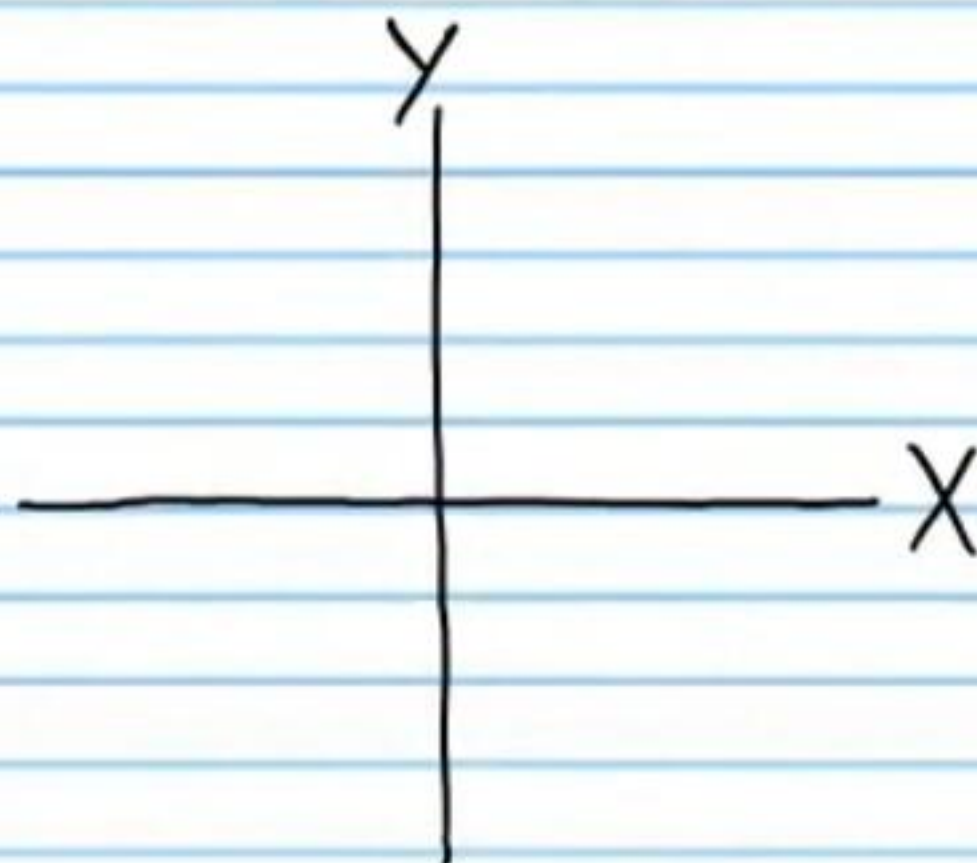
⑩ $\tan 74^\circ$



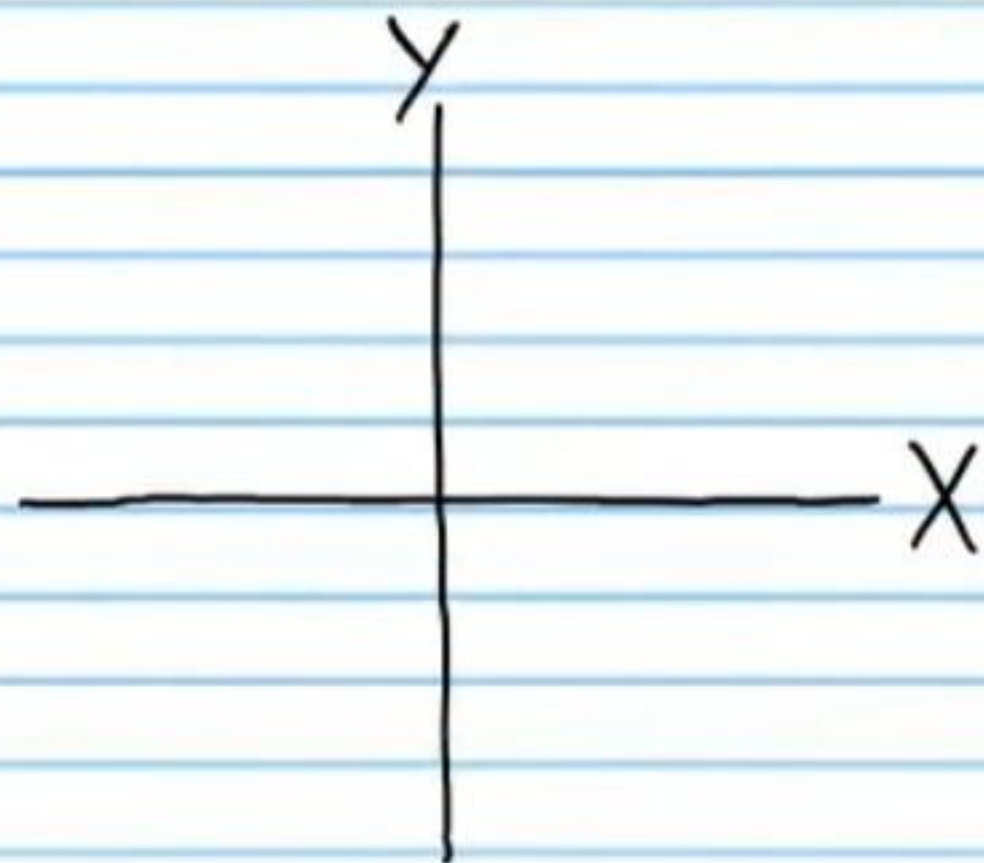
⑪ $\sin 160^\circ$



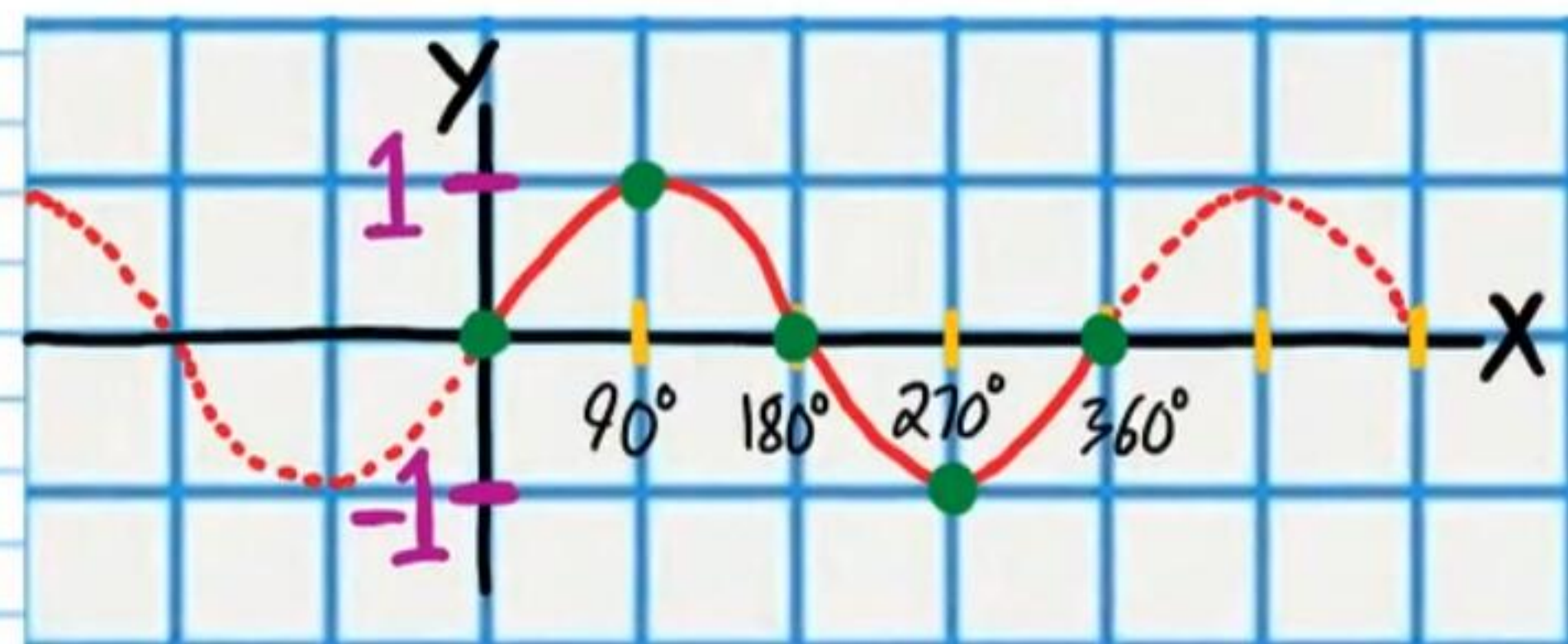
⑫ $\cos 251^\circ$



⑬ $\tan 318^\circ$



$y = \sin x$



Use the graph above to find the following values.

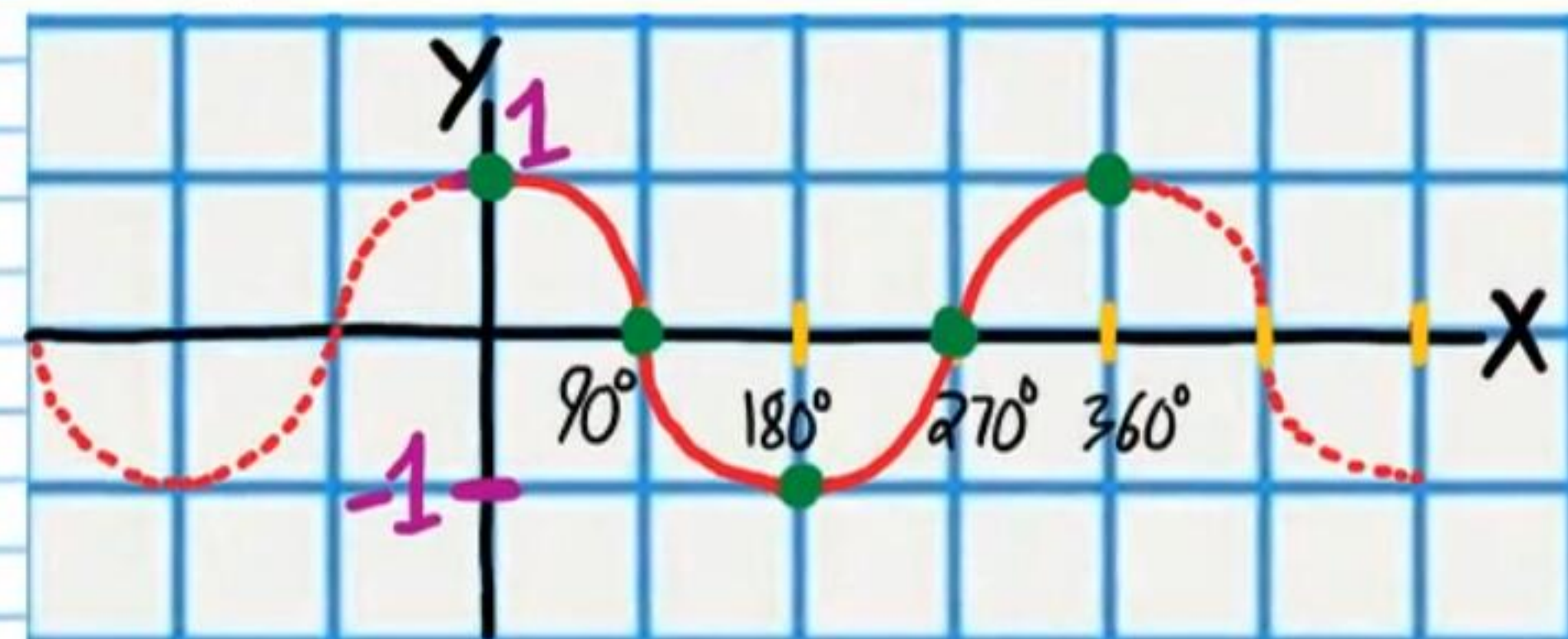
⑭ $\sin 90^\circ$

⑮ $\sin 180^\circ$

⑯ $\sin 540^\circ$

⑰ $\sin(-270^\circ)$

$y = \cos x$



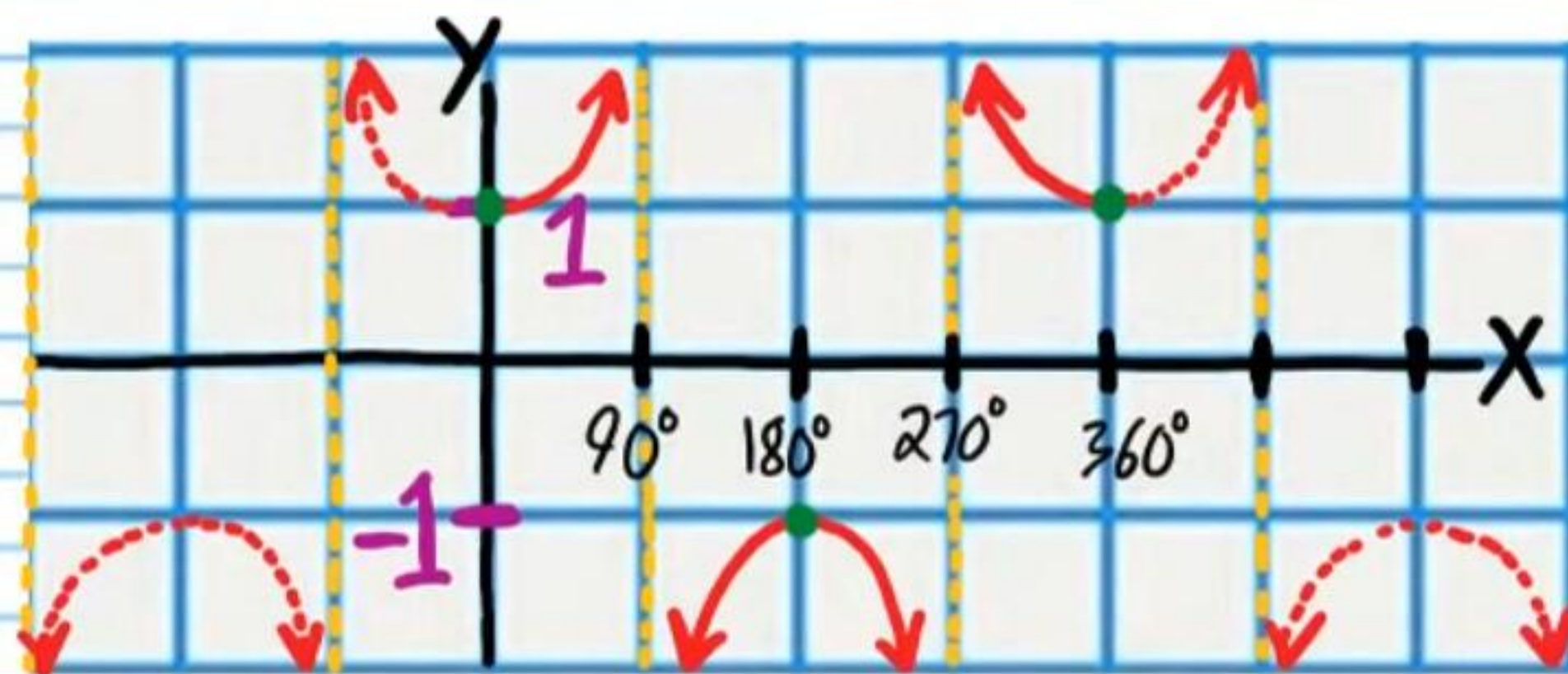
Use the graph above to find the following values.

⑱ $\cos 450^\circ$

⑲ $\cos(-180^\circ)$

⑳ $\cos(-270^\circ)$

$$y = \sec x$$

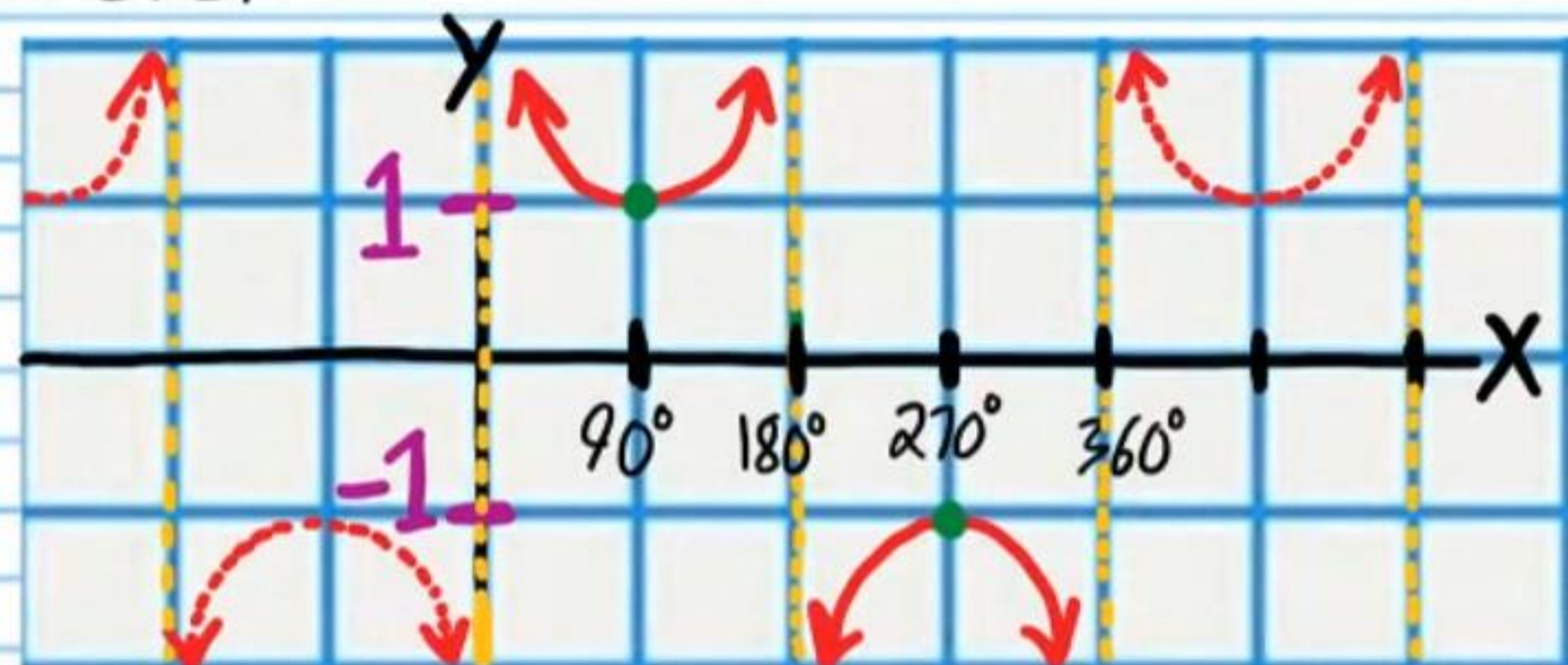


Use the graph above to find the following values.

②① $\sec(-270^\circ)$

②② $\sec 360^\circ$

$$y = \csc x$$

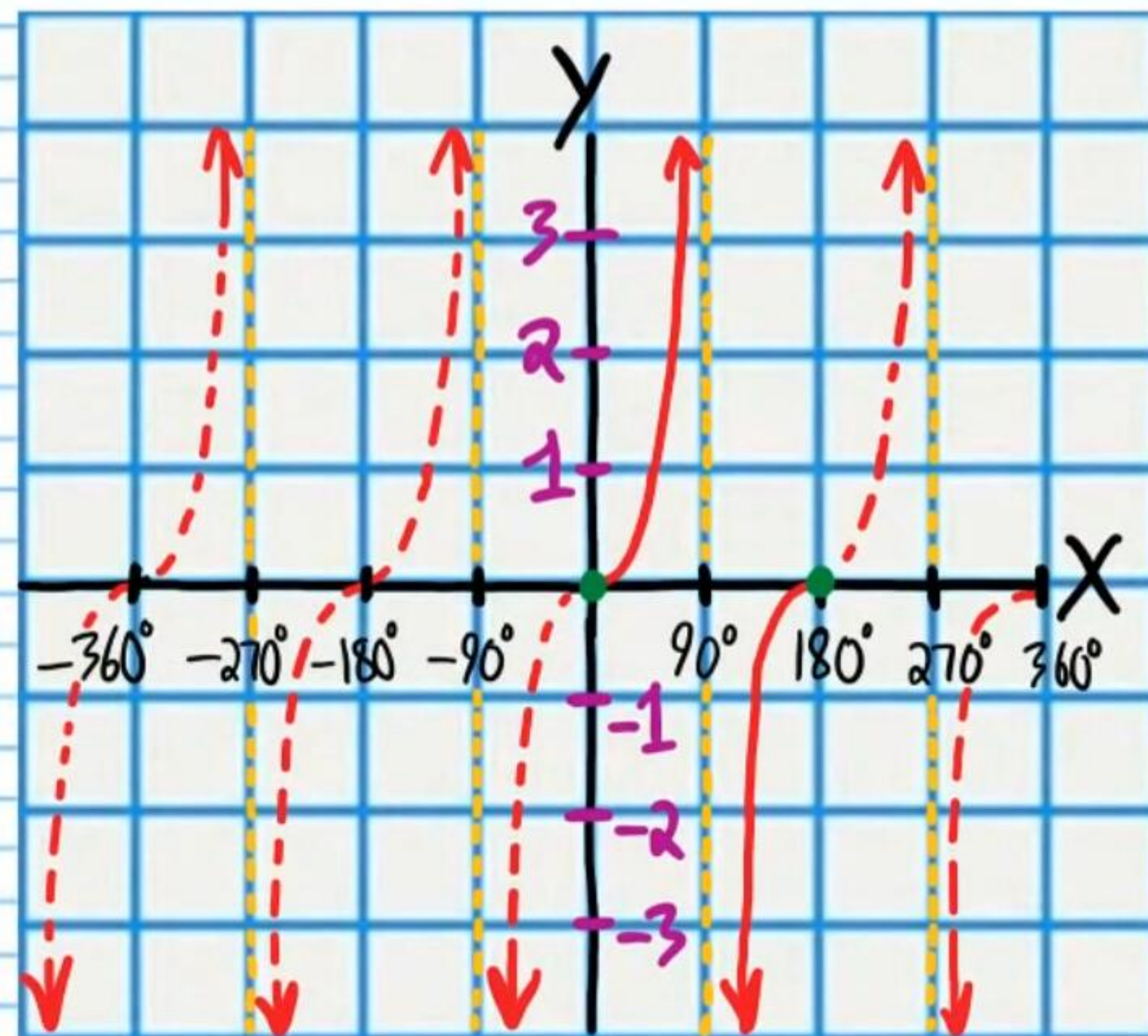


Use the graph above to find the following values.

②③ $\csc(-180^\circ)$

②④ $\csc 450^\circ$

$$y = \tan x$$

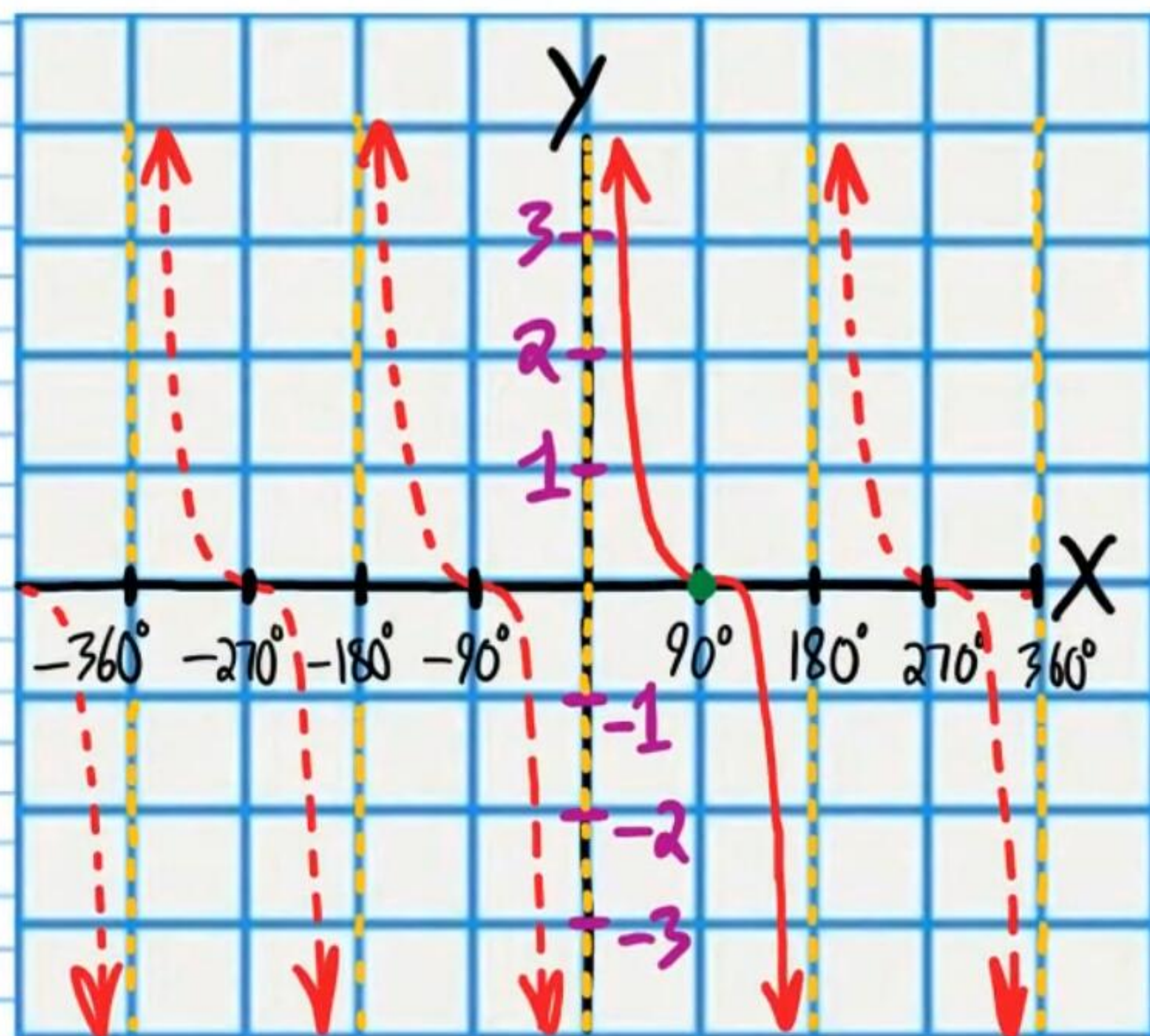


Use the graph above to find the following values.

②⑤ $\tan(-90^\circ)$

②⑥ $\tan 270^\circ$

$$y = \cot x$$



Use the graph above to find the following values.

(27) $\cot(-180^\circ)$

(28) $\cot(-270^\circ)$

The measures of angles in standard position are written below. Find the measures of three coterminal angles in standard position.

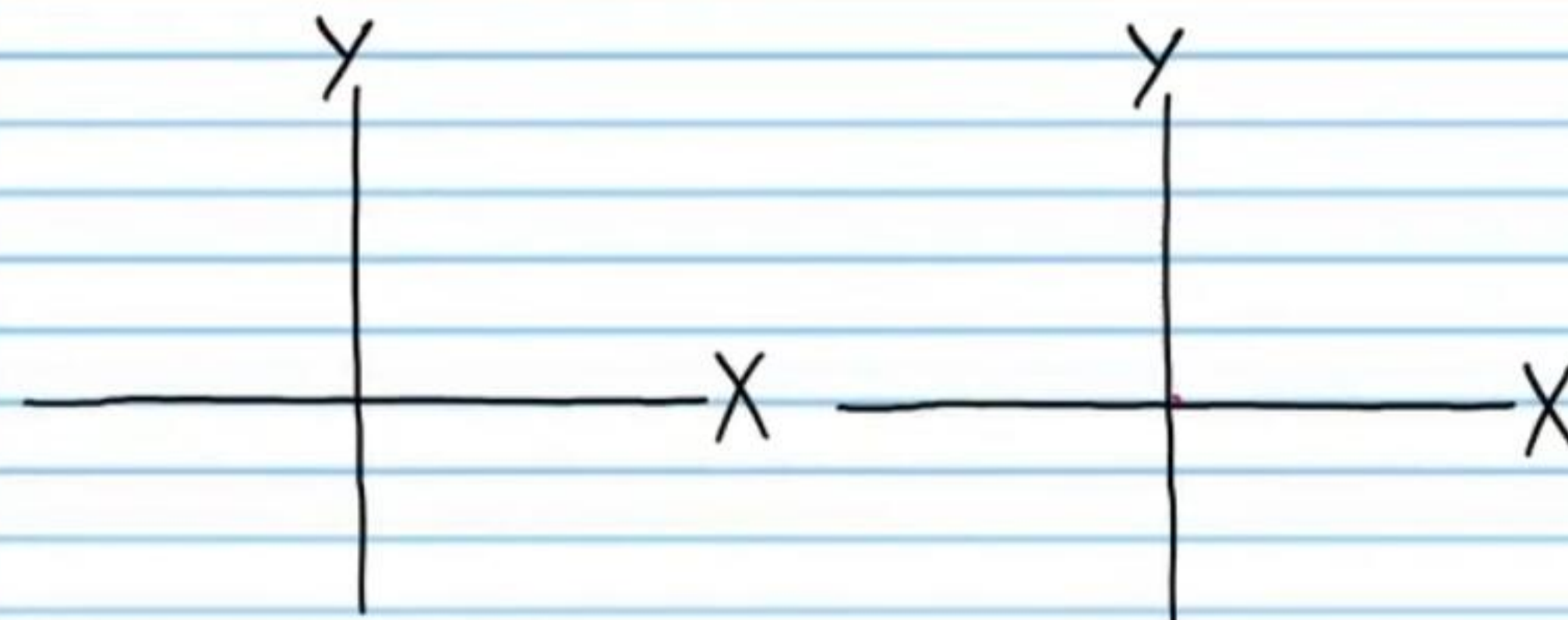
(29) 170°

(30) -42°

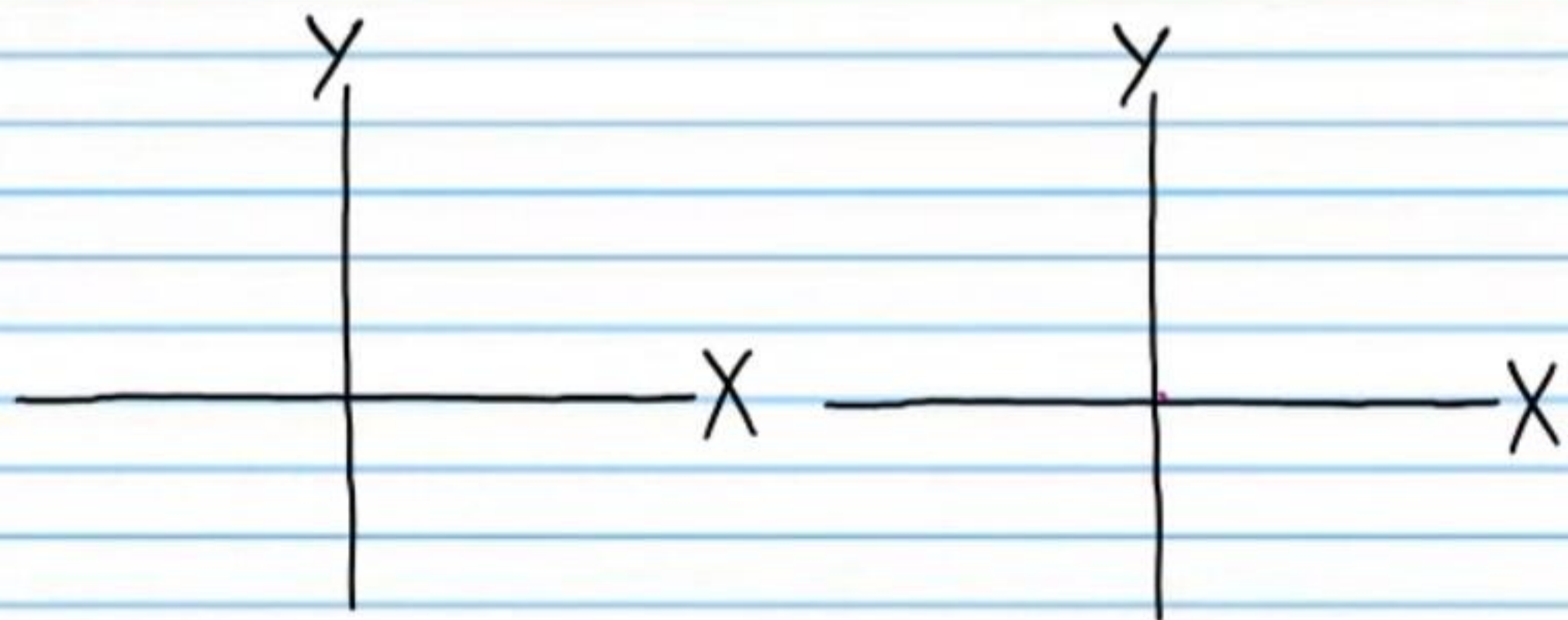
(31) 311°

Use a calculator to show that the following equations are true. Then, draw a reference triangle in standard position to show why each equation is true.

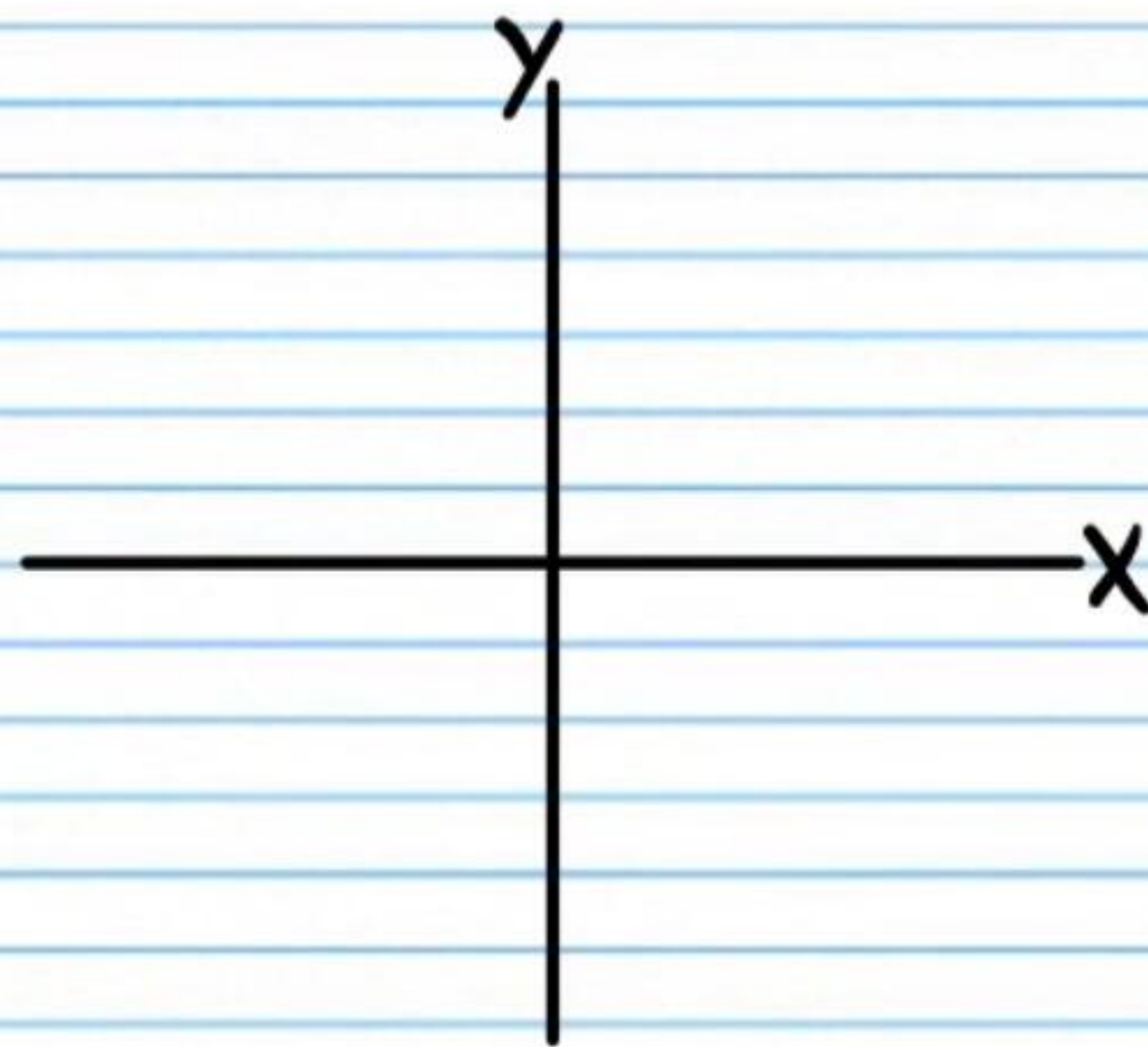
(32) $\sin 130^\circ = \sin 490^\circ$



③③ $\tan(-40^\circ) = \tan 320^\circ$



③④ If $\tan W = -\frac{1}{3}$, and W terminates in quadrant II, find the values of the five remaining basic trigonometric functions of W .



i) $\csc W$

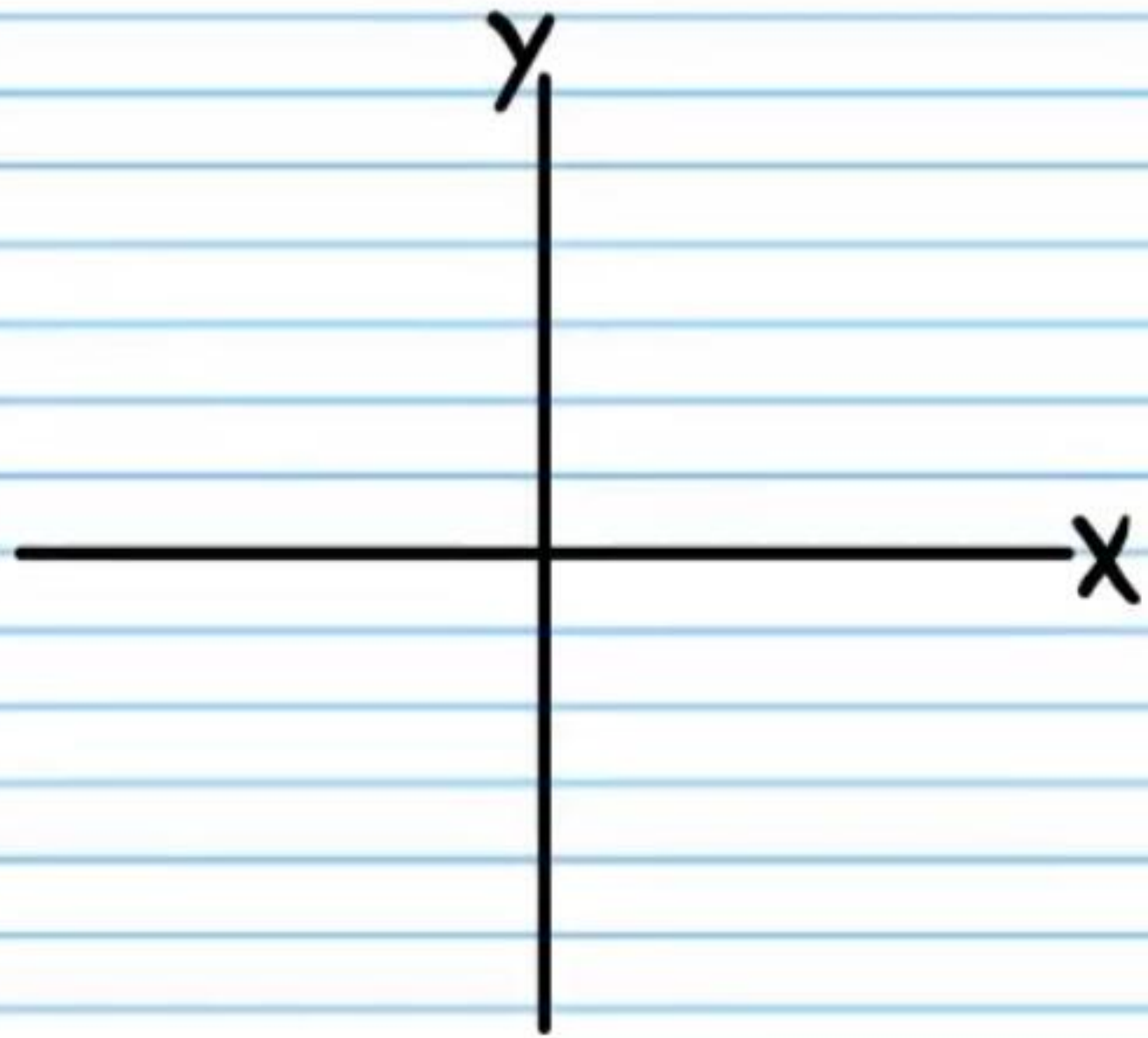
ii) $\sec W$

iii) $\cot W$

iv) $\cos W$

v) $\sin W$

35) If $\sec\theta = -\frac{5}{3}$, and θ terminates in quadrant III, find the values of the five remaining basic trigonometric functions of θ .



i) $\csc\theta$

ii) $\sec\theta$

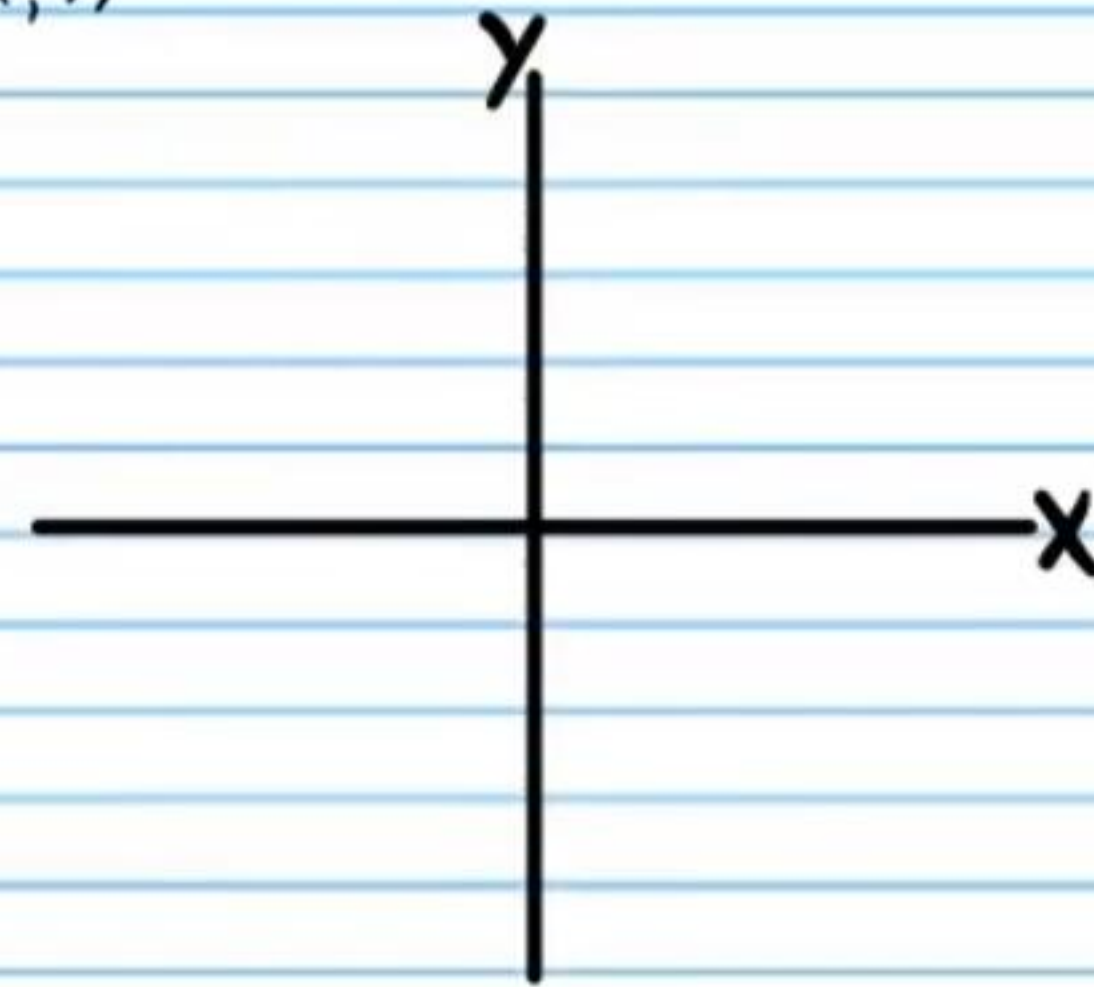
iii) $\cot\theta$

iv) $\cos\theta$

v) $\tan\theta$

The coordinates of points are given below. If each of these points lies on the terminal side of an angle in standard position, find the values of the six basic trigonometric functions of the angle.

36) (2, 3)



i) $\sin\theta$

ii) $\cos\theta$

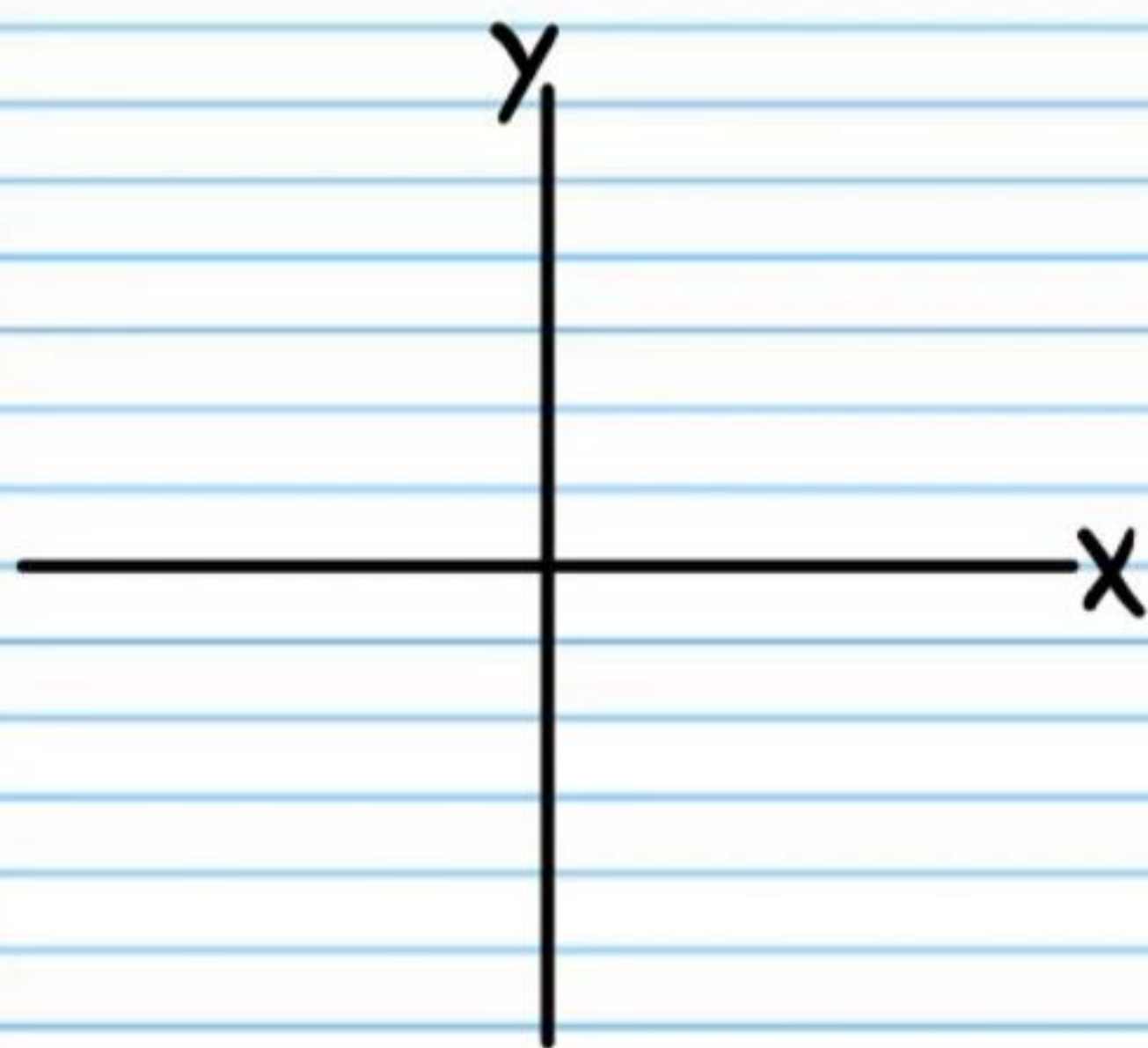
iii) $\tan\theta$

iv) $\csc\theta$

v) $\sec\theta$

vi) $\cot\theta$

③⑦ $(3, -3\sqrt{3})$



i) $\sin \theta$

ii) $\cos \theta$

iii) $\tan \theta$

iv) $\csc \theta$

v) $\sec \theta$

vi) $\cot \theta$

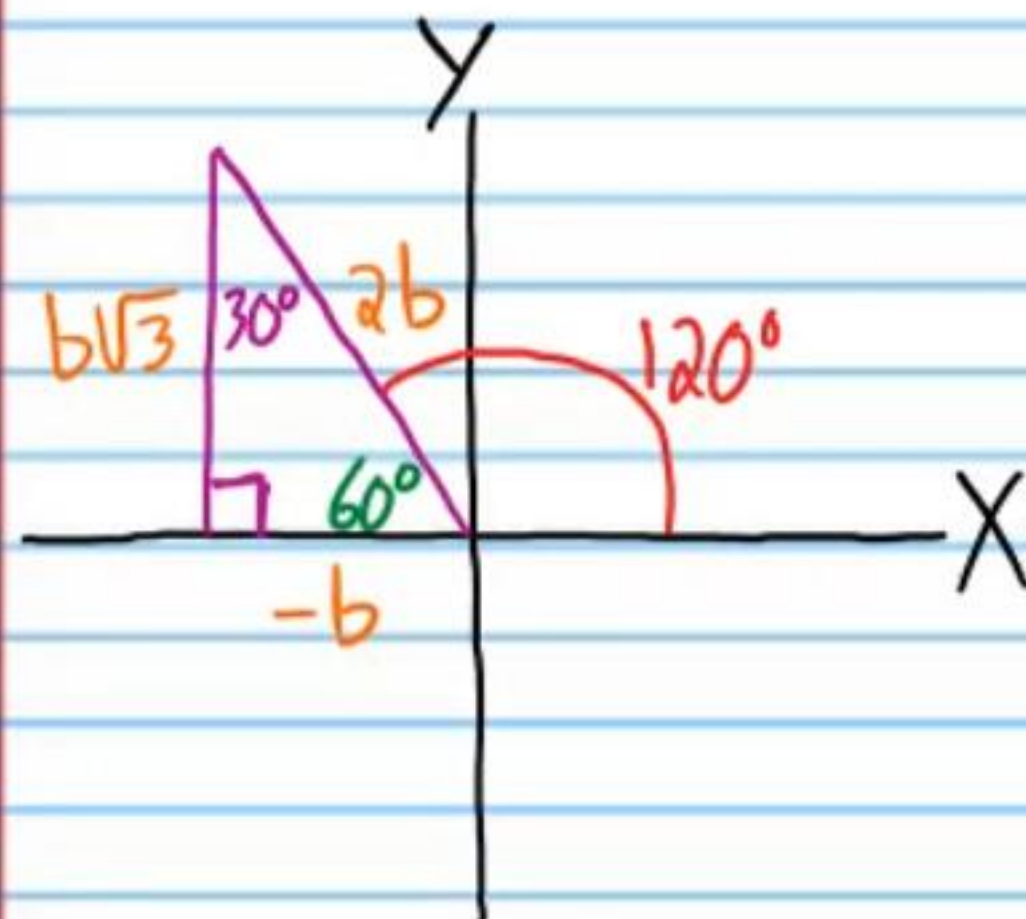
Homework Answers

① $\sin A = \frac{5}{13}$ $\cos \theta = \frac{12}{13}$ $\tan B = \frac{12}{5}$ $\csc \beta = \frac{13}{12}$ $\sec \theta = \frac{13}{12}$ $\cot A = \frac{12}{5}$

② $\sin \alpha = \frac{\sqrt{7}}{4}$ $\cos \beta = \frac{3}{4}$ $\tan A = \frac{3\sqrt{7}}{7}$ $\csc \alpha = \frac{4}{3}$ $\sec \alpha = \frac{4}{3}$ $\cot B = \frac{3\sqrt{7}}{7}$

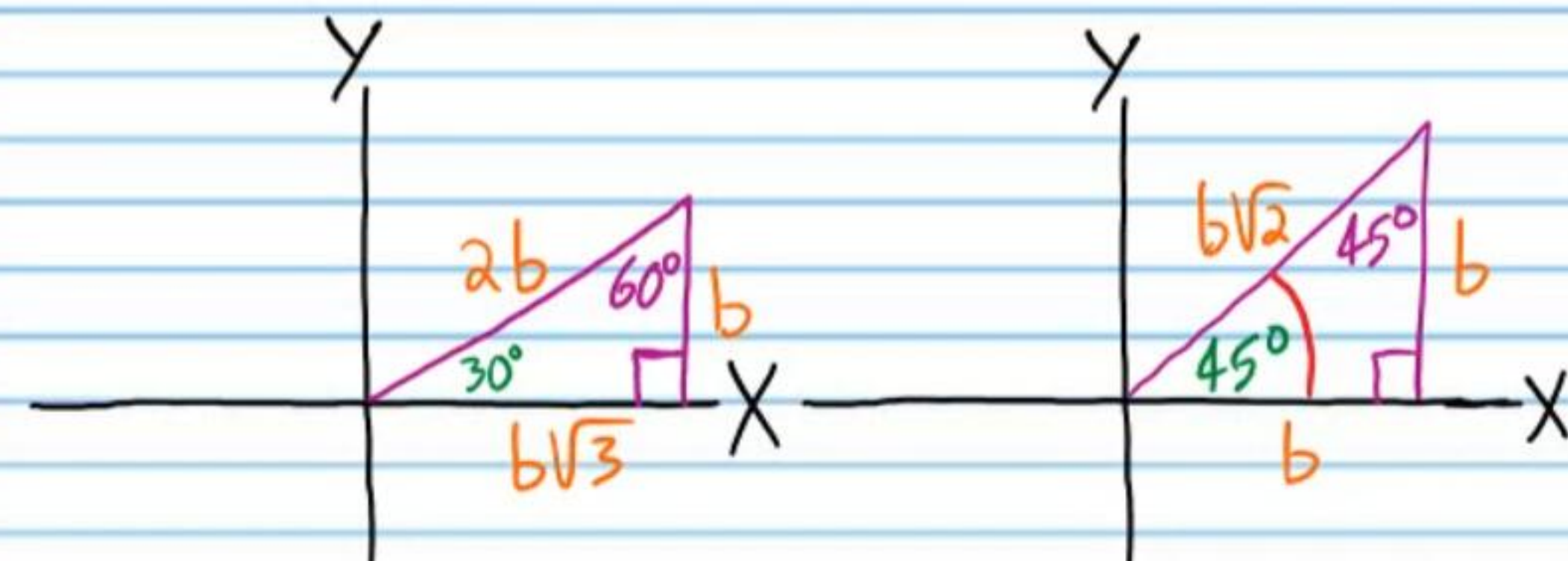
③ $\sin 50^\circ = \frac{x}{y}$ $\cos A = \frac{y}{r}$ $\tan 40^\circ = \frac{y}{x}$ $\csc B = \frac{r}{y}$ $\sec 50^\circ = \frac{r}{y}$ $\cot A = \frac{y}{x}$

④ $\tan 120^\circ = -\sqrt{3}$ ⑤ $\cos 225^\circ = -\frac{\sqrt{2}}{2}$

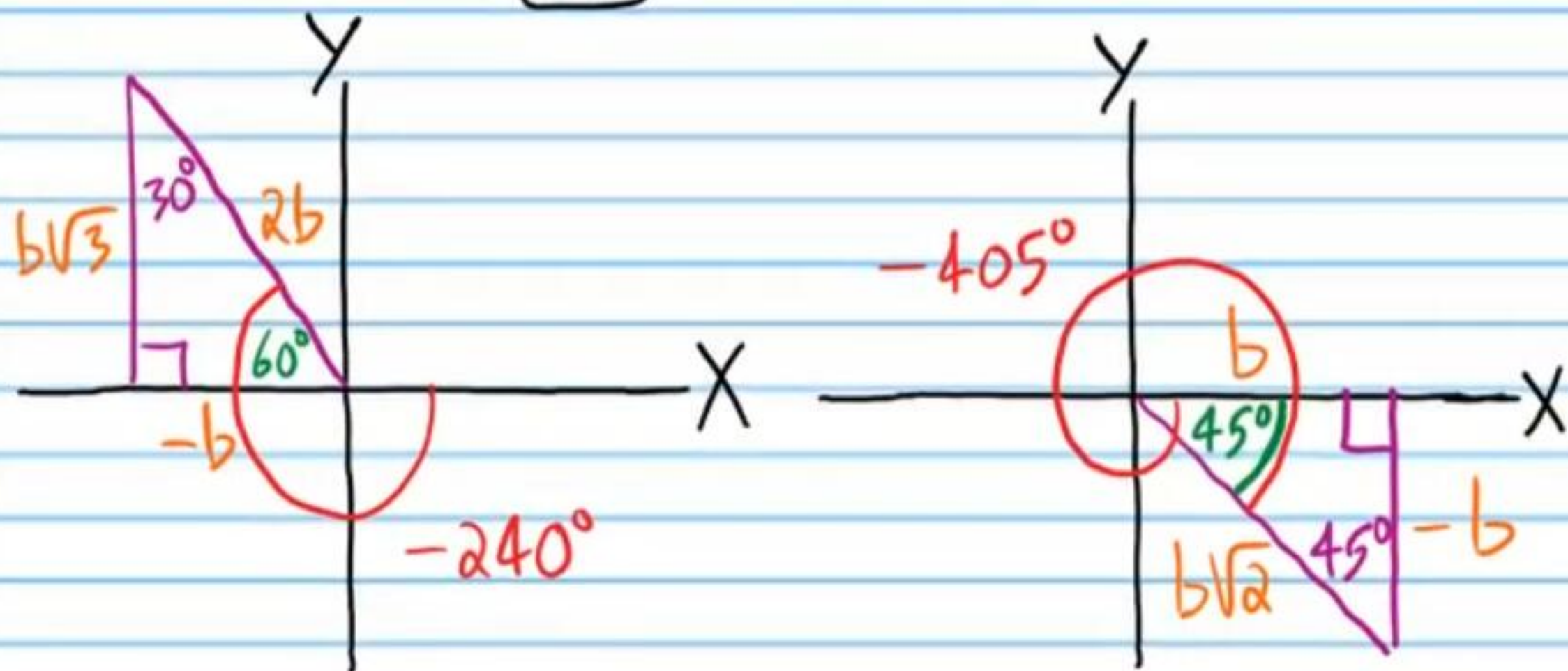


⑥ $\sin 30^\circ = \frac{1}{2}$

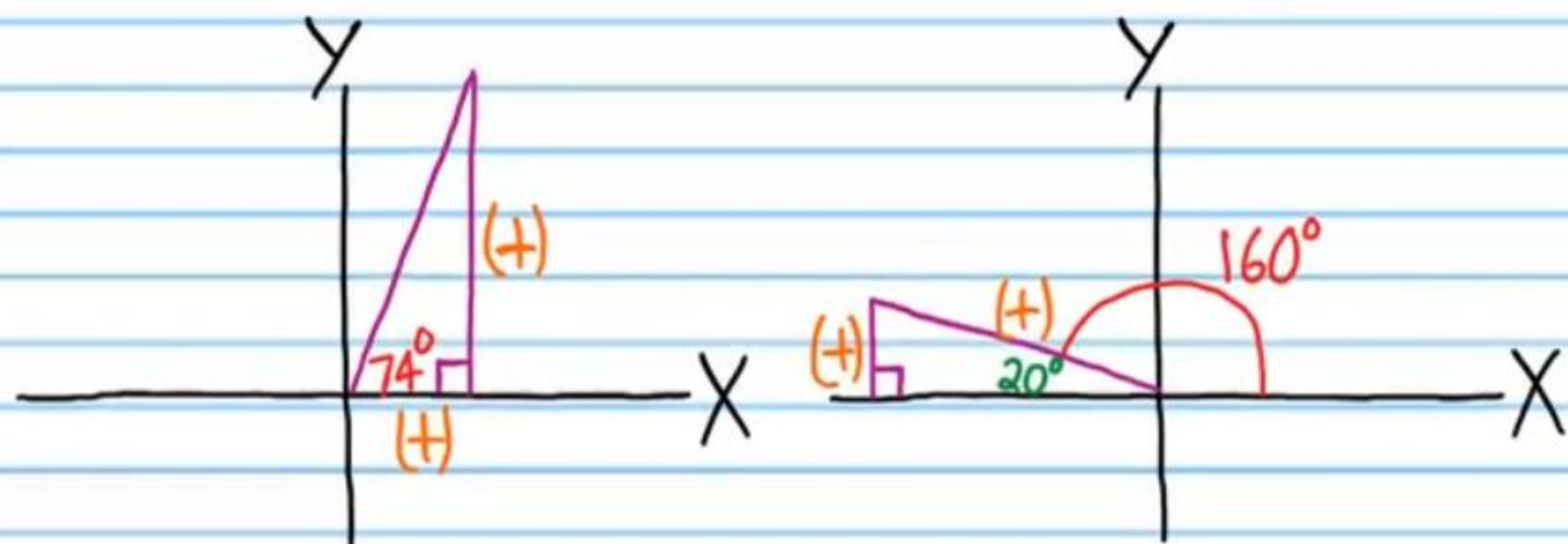
⑦ $\csc 45^\circ = \sqrt{2}$



⑧ $\csc(-240^\circ) = \boxed{\frac{2\sqrt{3}}{3}}$ ⑨ $\tan(-405^\circ) = \boxed{-1}$



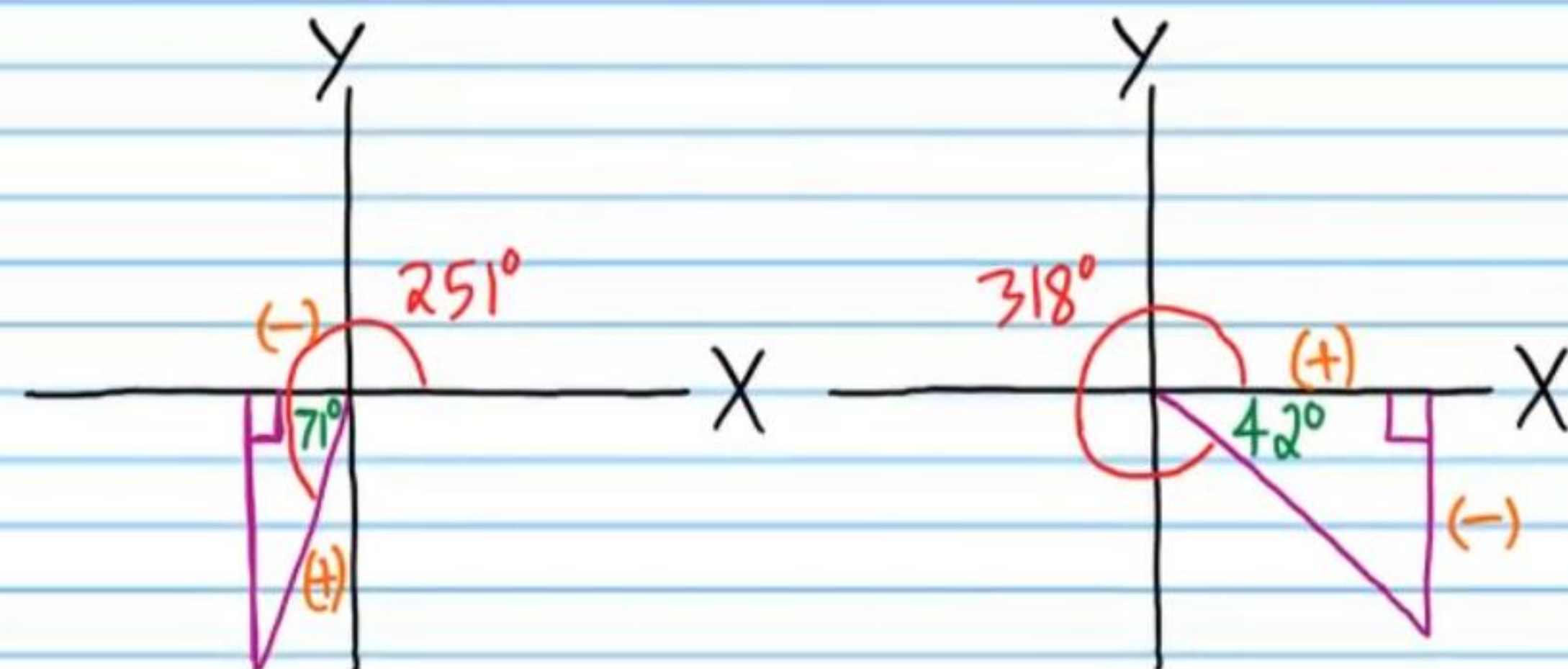
⑩ $\tan 74^\circ \approx \boxed{3.4874}$ ⑪ $\sin 160^\circ \approx \boxed{0.3420}$



$\frac{(+)}{(+)} = (+)$

$\frac{(+)}{(+)} = (+)$

⑫ $\cos 251^\circ \approx \boxed{-0.3256}$ ⑬ $\tan 318^\circ \approx \boxed{-0.9004}$



$\frac{(-)}{(+)} = (-)$

$\frac{(-)}{(+)} = (-)$

⑭ $\boxed{1}$ ⑮ $\boxed{0}$ ⑯ $\boxed{0}$ ⑰ $\boxed{1}$ ⑱ $\boxed{0}$ ⑲ $\boxed{-1}$ ⑳ $\boxed{0}$

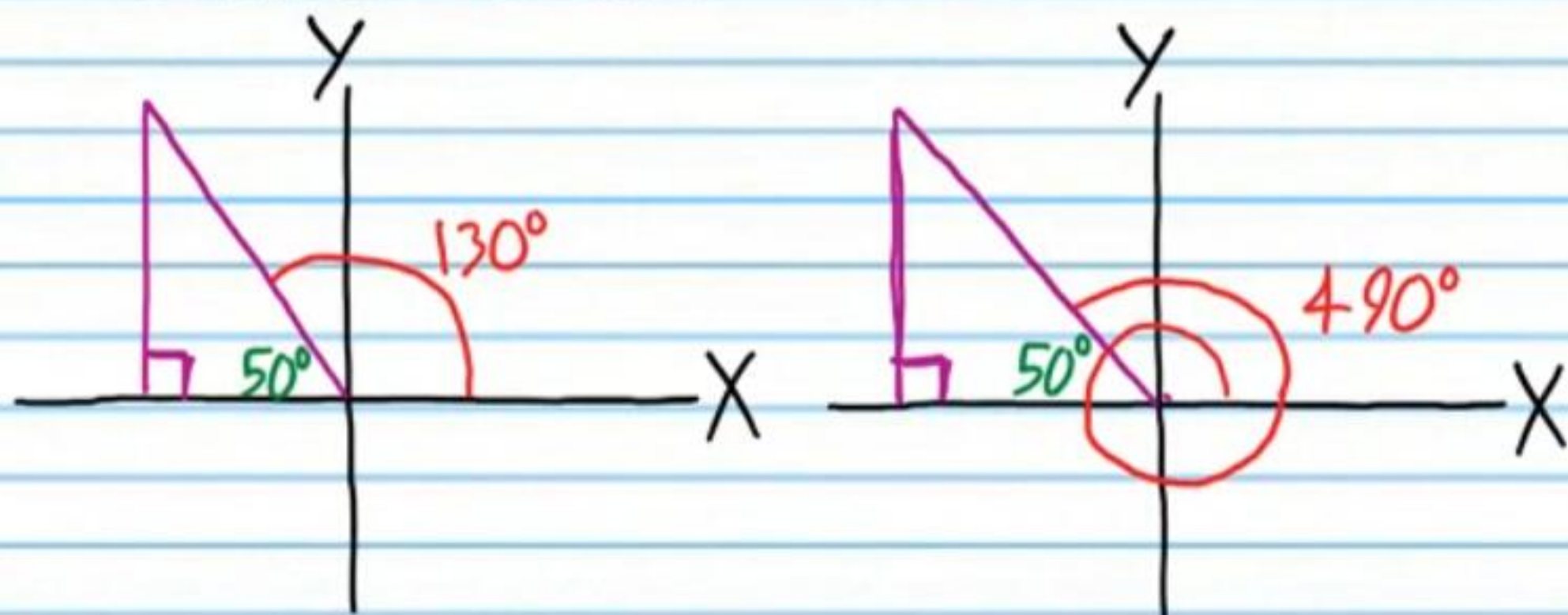
㉑ $\boxed{\text{Und}}$ ㉒ $\boxed{1}$ ㉓ $\boxed{\text{Und}}$ ㉔ $\boxed{1}$ ㉕ $\boxed{\text{Und}}$ ㉖ $\boxed{\text{Und}}$

㉗ $\boxed{\text{Und}}$ ㉘ $\boxed{0}$ Possible answers (only need three)

㉙ $\theta_1 = \boxed{-190^\circ}$ ㉚ $\theta_1 = \boxed{318^\circ}$ ㉛ $\theta_1 = \boxed{-49^\circ}$
 $\theta_2 = \boxed{530^\circ}$ $\theta_2 = \boxed{678^\circ}$ $\theta_2 = \boxed{-409^\circ}$
 $\theta_3 = \boxed{-550^\circ}$ $\theta_3 = \boxed{-402^\circ}$ $\theta_3 = \boxed{671^\circ}$
 $\theta_4 = \boxed{890^\circ}$ $\theta_4 = \boxed{-762^\circ}$ $\theta_4 = \boxed{1,031^\circ}$
 $\theta_5 = \boxed{-769^\circ}$

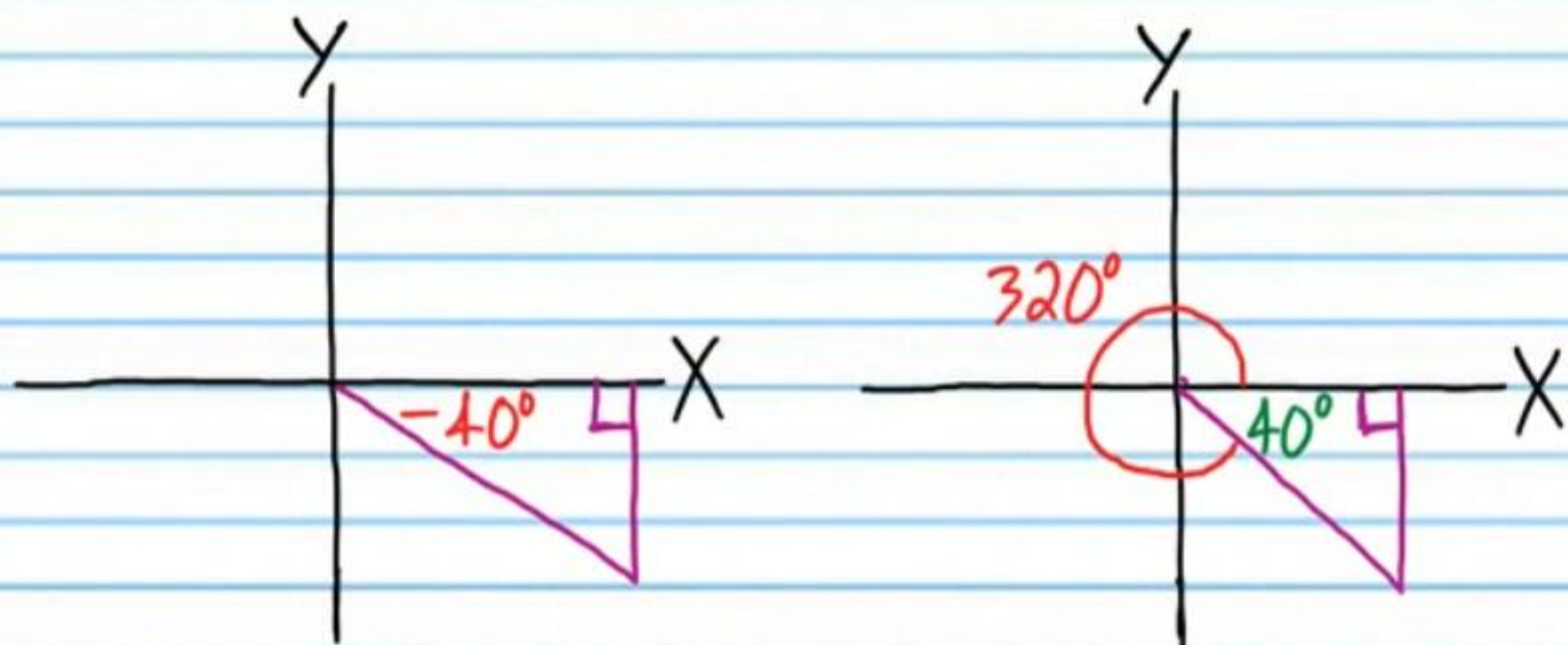
$$\textcircled{32} \sin 130^\circ = \sin 490^\circ$$

$$0.7660 = 0.7660$$



$$\textcircled{33} \tan(-40^\circ) = \tan 320^\circ$$

$$-0.8391 = -0.8391$$



$$\textcircled{34} \text{ i) } \csc w = \boxed{\sqrt{10}}$$

$$\text{ii) } \sec w = \boxed{-\frac{\sqrt{10}}{3}}$$

$$\text{iii) } \cot w = \boxed{-3}$$

$$\text{iv) } \cos w = -\frac{3}{\sqrt{10}} = \boxed{-\frac{3\sqrt{10}}{10}}$$

$$\text{v) } \sin w = \frac{1}{\sqrt{10}} = \boxed{\frac{\sqrt{10}}{10}}$$

$$\textcircled{35} \text{ i) } \csc \theta = \boxed{-\frac{5}{4}}$$

$$\text{ii) } \sec \theta = \boxed{-\frac{5}{3}}$$

$$\text{iii) } \cot \theta = \boxed{\frac{3}{4}}$$

$$\text{iv) } \cos \theta = \boxed{-\frac{3}{5}}$$

$$\text{v) } \tan \theta = \boxed{\frac{4}{3}}$$

$$\textcircled{36} \text{ i) } \sin \theta = \boxed{\frac{3\sqrt{13}}{13}}$$

$$\text{ii) } \cos \theta = \boxed{\frac{2\sqrt{13}}{13}}$$

$$\text{iii) } \tan \theta = \boxed{\frac{3}{2}}$$

$$\text{iv) } \csc \theta = \boxed{\frac{\sqrt{13}}{3}}$$

$$\text{v) } \sec \theta = \boxed{\frac{\sqrt{13}}{2}}$$

$$\text{vi) } \cot \theta = \boxed{\frac{2}{3}}$$

$$\textcircled{37} \text{ i) } \sin \theta = \boxed{-\frac{\sqrt{3}}{2}}$$

$$\text{ii) } \cos \theta = \boxed{\frac{1}{2}}$$

$$\text{iii) } \tan \theta = \boxed{-\sqrt{3}}$$

$$\text{iv) } \csc \theta = \boxed{-\frac{2\sqrt{3}}{3}}$$

$$\text{v) } \sec \theta = \boxed{2}$$

$$\text{vi) } \cot \theta = \boxed{-\frac{\sqrt{3}}{3}}$$